# Chapter 9: Random Processes – Part I

## 9.1 & 9.2 Definition and Specification of a Stochastic Process

9.1

We find the probabilities of the events $\{X_1 = i, X_2 = j\}$ in terms of the probabilities of the equivalent events of $\xi$:

$$\begin{aligned} P[X_1 = 1, X_2 = 1] &= P\left[\frac{3}{4} < \xi < 1\right] = \frac{1}{4} \\ P[X_1 = 0, X_2 = 1] &= P\left[\frac{1}{4} < \xi < \frac{1}{2}\right] = \frac{1}{4} \\ P[X_1 = 1, X_2 = 0] &= P\left[\frac{1}{2} < \xi < \frac{3}{4}\right] = \frac{1}{4} \\ P[X_1 = 0, X_2 = 0] &= P\left[0 < \xi < \frac{1}{4}\right] = \frac{1}{4} \end{aligned}$$

$\Rightarrow P[X_1 = i, X_2 = j] = P[X_1 = i]P[X_2 = j]$ all $i, j \in \{0, 1\}$
$\Rightarrow X_1, X_2$ independent RV's

9.2

a) $\xi$ is the output of fair die

if $\xi=1$ : $X_n$ : ··· 1111 ···

$\xi=2$ : $X_n$ : ··· 2222 ···

⋮

$\xi=6$ : $X_n$ : ··· 6666 ···

b) $P[X_n=1] = P\{k=1\} = 1/6$

$P[X_n=2] = P\{k=2\} = 1/6$

⋮

$P[X_n=6] = P\{k=6\} = 1/6$

$P[X_n=k] = \emptyset \quad \forall k \notin \{1,2,3,4,5,6\}$

c) $P[X_n=1, X_{n+k}=1] = \frac{1}{6}$

$P[X_n=2, X_{n+k}=2] = 1/6$

⋮

$P[X_n=6, X_{n+k}=6] = 1/6$

$P[X_n=k_1, X_{n+k}=k_2] = 0 \quad \forall k_1 \neq k_2$

$P[X_n=k_1, X_{n+k}=k_2] = 0 \quad \forall k_1 \notin \{1,2,3,4,5,6\}$

d) $E[X_n] = 1\times P\{X_n=1\} + 2\times P\{X_n=2\} + 3\times P\{X_n=3\} + \cdots + 6\times P\{X_n=6\} = \frac{21}{6}$

$E[X_n X_{n+k}] = 1\times 1\times P\{X_n=1, X_{n+k}=1\} + 2\times 2\times P\{X_n=2, X_{n+k}=2\} + \cdots + 6\times 6\times P\{X_n=6, X_{n+k}=6\} = \frac{91}{6}$

$C_X(n,n+k) = E[X_n X_{n+k}] - E[X_n]E[X_{n+k}] = \frac{91}{6} - \frac{21}{6}\times\frac{21}{6} = 2.9167$

9.3

~~6.3~~ **a)**

| | | ... | $n=0$ | $n=1$ | $n=2$ | ... | |
|---|---|---|---|---|---|---|---|
| If $\xi =$ Heads | $X_n$ | ... | 1 | $-1$ | 1 | $-1$ | ... |
| If $\xi =$ Tails | $X_m$ | ... | $-1$ | 1 | $-1$ | 1 | ... |

**b)** $n$ even $\quad P[X_n = 1] = P[\text{Heads}] = \frac{1}{2}$

$n$ odd $\quad P[X_m = 1] = P[\text{Tails}] = \frac{1}{2}$

**c)** $k$ even

$$\begin{aligned} P[X_n = 1, X_{n+k} = 1] &= P[\text{Heads}] = \frac{1}{2} \\ P[X_n = -1, X_{n+k} = -1] &= P[\text{Tails}] = \frac{1}{2} \\ P[X_n = \pm 1, X_{n+k} = \mp 1] &= 0 \end{aligned}$$

$k$ odd

$$\begin{aligned} P[X_n = 1, X_{n+k} = -1] &= P[\text{Heads}] = \frac{1}{2} \\ P[X_n = -1, X_{n+k} = 1] &= P[\text{Tails}] = \frac{1}{2} \\ P[X_n = \pm 1, X_{n+k} = \pm 1] &= 0 \end{aligned}$$

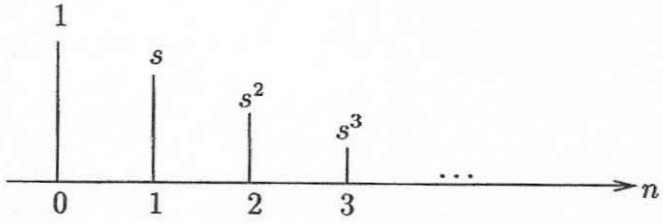
**d)** $\mathcal{E}[X_n] = 1\left(\frac{1}{2}\right) + (-10)\frac{1}{2} = 0$

$k$ even $\quad \mathcal{E}[X_n X_{n+k}] = (1)^2\frac{1}{2} + (-1)^2\frac{1}{2} = 1$

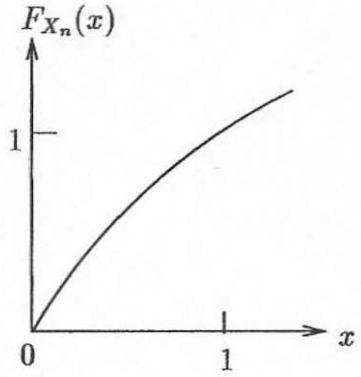
$k$ odd $\quad \mathcal{E}[X_n X_{n+k}] = (1)(-1)\frac{1}{2} + (-1)(1)\frac{1}{2} = -1$

9.4

**6.4 a)** $X_n = s^n \qquad 0 < s < 1$




**b)** $P[X_n \leq x] = P[s^n \leq x] = P[s \leq x^{1/n}] = x^{1/n} \qquad 0 \leq x \leq 1$




**c)** For $0 < x, y < 1$

$$\begin{aligned} P[X_n \leq x, X_{n+1} \leq y] &= P[s^n \leq x, s^{n+1} \leq y] \\ &= P[s \leq x^{1/n}, s \leq y^{1/n+1}] \\ &= P[s \leq \min(x^{1/n}, y^{1/n+1})] \\ &= \min(x^{1/n}, y^{1/n+1}) \end{aligned}$$

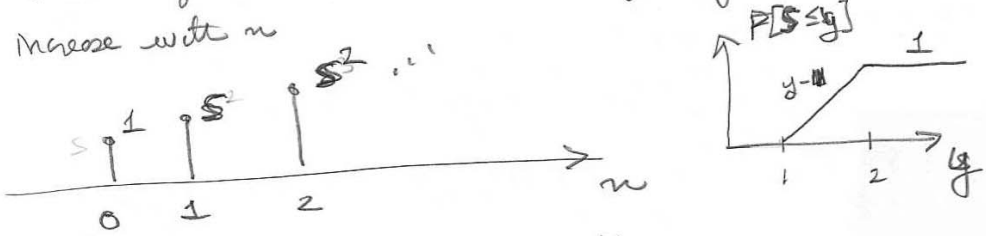
**d)** $\mathcal{E}[X_n] = \mathcal{E}[s^n] = \int_0^1 s^n ds = \dfrac{1}{n+1}$

$\mathcal{E}[X_n X_{n+k}] = \mathcal{E}[s^n s^{n+k}] = \mathcal{E}[s^{2n+k}] = \frac{1}{2n+k+1}$

$C_X(n, n+k) = \frac{1}{2n+k+1} - \left(\frac{1}{n+1}\right)\left(\frac{1}{n+k+1}\right)$

— continued —

(9.4) (c) If $S$ is uniform in $(1,2)$ then the sample paths increase with $n$




$$P[X_n \le x] = P[S^n \le x] = P[S \le x^{1/n}] \qquad 1 < x^{1/n} < 2$$

$$= x^{1/n} - 1$$

$$P[X_n \le x, X_{n+1} \le y] = P[S^n \le x, S^{n+1} \le y]$$

$$= P[S \le \min(x^{1/n}, y^{1/(n+1)})]$$

$$= \min(x^{1/n}, y^{\frac{1}{n+1}}) - 1$$

$$E[X_n] = E[S^n] = \int_1^2 s^n ds = \frac{2^{n+1} - 1}{n+1}$$

$$E[X_n X_{n+k}] = E[S^{2n+k}] = \frac{2^{2n+k+1} - 1}{2n+k+1}$$

$$C_X(n, n+k) = \frac{2^{2n+k+1} - 1}{2n+k+1} - \left(\frac{2^{n+1} - 1}{n+1}\right)\left(\frac{2^{n+k+1} - 1}{n+k+1}\right)$$

9.5

**6.5 a)** Since $g(t)$ is zero outside the interval [0,1]:

$$P[X(t) = 0] = 1 \text{ for } t \notin [0,1]$$

For $t \in [0,1]$, we have

$$P[X(t) = 1] = P[X(t) = -1] = \frac{1}{2}$$

**b)** $$m_X(t) = \begin{cases} 1 \cdot P[X(t) = 1] + (-1)P[X(t) = -1] = 0 & 0 \le t \le 1 \\ 0 & \text{otherwise} \end{cases}$$

**c)** For $t \in [0,1]$, $t + d \in [0,1]$, $X(t)$ must be the same value, thus:

$$\begin{aligned} P[X(t) = \pm 1, X(t+d) = \pm 1] &= \frac{1}{2} \\ P[X(t) = \pm 1, X(t+d) = \mp 1] &= 0 \end{aligned}$$

For $t \in [0,1]$, $t + d \notin [0,1]$:

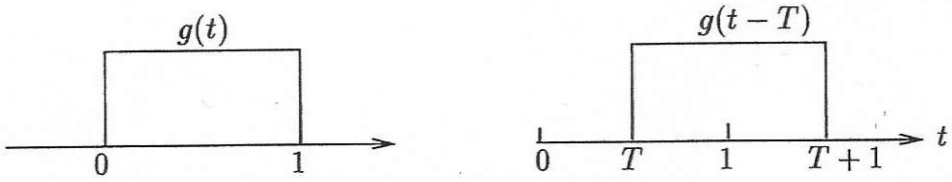
$$P[X(t) = \pm 1, X(t+d) = 0] = \frac{1}{2}$$

For $t \notin [0,1]$, $t + d \notin [0,1]$:

$$P[X(t) = 0, X(t+d) = 0] = 1$$

**d)**
$$\begin{aligned} C_X(t, t+d) &= \mathcal{E}[X(t)X(t+d)] - m_X(t)m_X(t+d) \\ &= \mathcal{E}[X(t)X(t+d)] \\ &= \begin{cases} 1 & t \in [0,1] \text{ and } t+d \in [0,1] \\ 0 & \text{otherwise} \end{cases} \end{aligned}$$

9.6




$$P[Y(t) = 1] = P[g(t - T) = 1]$$

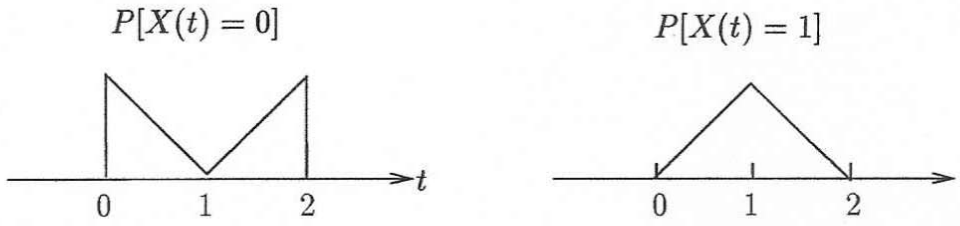
**a)**

For $0 < t < 1$

$$P[Y(t) = 0] = P[t < T] = 1 - t = 1 - P[Y(t) = 1]$$

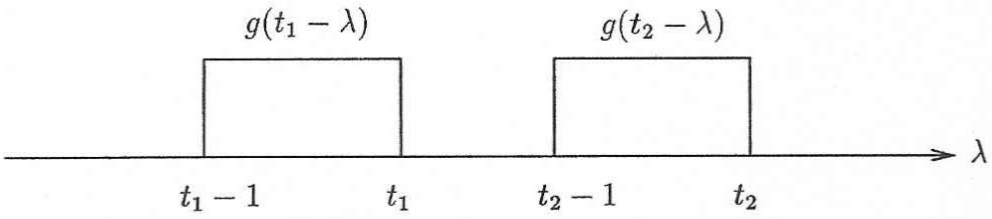
For $1 < t < 2$

$$\begin{aligned} P[Y(t) = 1] &= P[t < T + 1] = P[T > t - 1] = 1 - (t - 1) \\ &= 2 - t \\ P[Y(t) = 0] &= 1 - P[Y(t) = 1] = t - 1 \end{aligned}$$




**b)** $\mathcal{E}[Y(t)] = 1 \cdot P[Y(t) = 1] = P[Y(t) = 1]$

$$\begin{aligned} E[Y(t_1)Y(t_2)] &= \int_0^1 E[g(t_1 - T)g(t_2 - T)|T = \lambda] f_T(\lambda) d\lambda \\ &= \int_0^1 g(t_1 - \lambda)g(t_2 - \lambda) d\lambda \end{aligned}$$



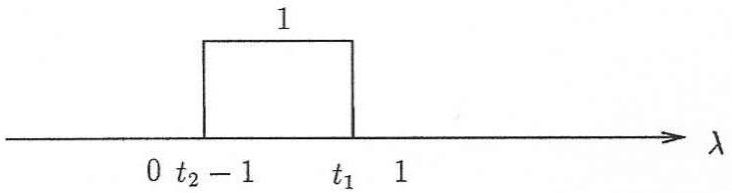

$$g(t_1 - \lambda)g(t_2 - \lambda) = 0 \quad \text{for} \quad t_1 < t_2 - 1$$

$$\Rightarrow \; R_Y(t_1, t_2) = 0 \quad \text{for} \quad t_2 - t_1 > 1$$

If $t_2 - 1 < t$, then

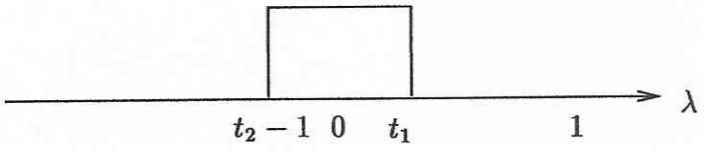
$$g(t_1 - \lambda)g(t_2 - \lambda) = \begin{cases} 1 & t_2 - 1 < \lambda < t_1 \\ 0 & \text{elsewhere} \end{cases}$$

**Case 1**



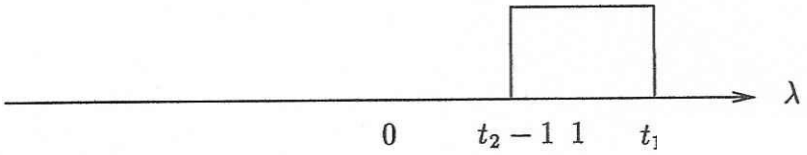

$$R_Y(t_1, t_2) = t_1 - (t_2 - 1) = 1 - (t_2 - t_1) \qquad t_1 < 1, \;\; 0 < t_2 - 1, \;\; t_2 - t_1 < 1$$

**Case 2**




$$R_Y(t_1, t_2) = t_1 \qquad t_1 < 1, \quad t_2 < 1, \quad t_2 - t_1 < 1$$

**Case 3**




$$R_Y(t_1, t_2) = 2 - t_2 \qquad t_1 > 1, \quad t_2 < 2$$

9.7 a) We will use conditional probability:

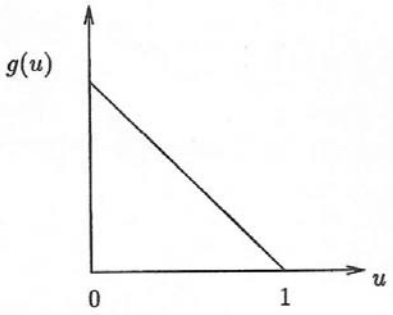
$$\begin{aligned} P[X(t) \leq x] &= P[g(t-T) \leq x] \\ &= \int_0^1 P[g(t-T) \leq x | T = \lambda] f_T(\lambda) d\lambda \\ &= \int_0^1 P[g(t-\lambda) \leq x] d\lambda \quad \text{since } f_T(\lambda) = 1 \\ &= \int_{t-1}^t P[g(u) \leq x] du \quad \text{after letting } u = t - \lambda \end{aligned}$$

$g(u)$ (and hence $P[g(u) \leq x]$) is a periodic function of $u$ with period, so we can change the limits of the above integral to any full period. Thus

$$P[X(t) \leq x] = \int_0^1 P[g(u) \leq x] du$$

Note that $g(u)$ is deterministic, so

$$P[g(u) \leq x] = \begin{cases} 1 & u : g(u) \leq x \\ 0 & u : g(u) > x \end{cases}$$




So finally

$$P[X(t) \leq x] = \int_{u:g(u)\leq x} 1 \, du = \int_{1-X}^{1} 1 \, du = x \, .$$

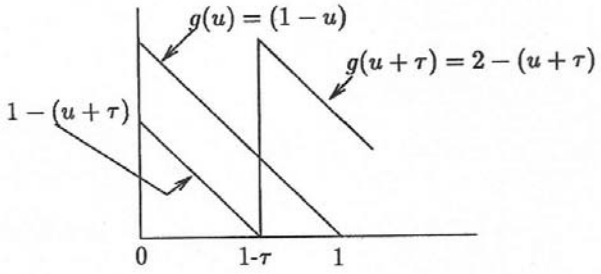
b) $m_X(t) = E[X(t)] = \int_0^1 x \, dx = \frac{1}{2}$.

The correlation is again found using conditioning on $T$:

$$\begin{aligned} E[X(t)X(t+\tau)] &= \int_0^1 E[g(t-T)g(t+\tau-T)|T=\lambda]f_T(\lambda)d\lambda \\ &= \int_0^1 g(t-\lambda)g(t+\tau-\lambda)d\lambda \\ &= \int_{t-1}^{t} g(u)g(u+\tau)du \end{aligned}$$

$g(u)g(u+\tau)$ is a periodic function in $u$ so we can change the limits to (0,1):

$$E[X(t)X(t+\tau)] = \int_0^1 g(u)g(u+\tau)du$$



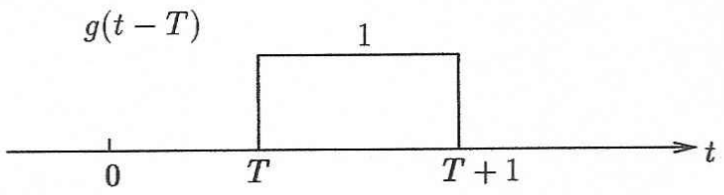

here we assume $0 < \tau < 1$ since $E[X(t)X(t+\tau)]$ is periodic in $\tau$.

$$\begin{aligned} E[X(t)X(t+\tau)] &= \int_0^{1-\tau}(1-u)(1-u-\tau)du + \int_{1-\tau}^{1}(1-u)(2-u-\tau)du \\ &= \frac{1}{3} - \frac{\tau}{2} + \frac{\tau^3}{6} + \frac{\tau^2}{2} - \frac{\tau^3}{6} \\ &= \frac{1}{3} - \frac{\tau}{2} + \frac{\tau^2}{2} \quad . \end{aligned}$$

Thus

$$\begin{aligned} C_X(t,t+\tau) &= \frac{1}{3} - \frac{\tau}{2} + \frac{\tau^2}{2} - \frac{1}{4} \\ &= \frac{1}{12} - \frac{\tau}{2} + \frac{\tau^2}{2} \end{aligned}$$

9.8

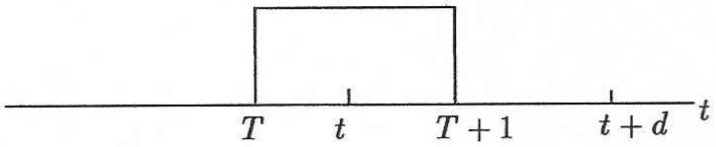



**a)**

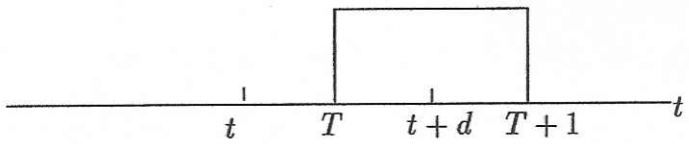
$$\begin{aligned} P[Y(t)=1] &= P[T<t<T+1] \\ &= P[t-1<T<t] = e^{-\alpha(t-1)} - e^{-\alpha t} \\ P[Y(t)=0] &= 1 - P[Y(t)=1] \end{aligned}$$

**b) Case 1** $d > 1$

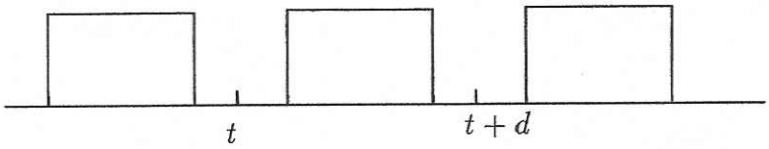
$$P[Y(t)=1, Y(t+d)=1] = 0$$



$$\begin{aligned}
&P[Y(t)=1, Y(t+d)=0] \\
&\quad = P[\{T<t<T+1\} \cap \{T+1<t+d\}] \\
&\quad = P[\{t-1<T<t\} \cap \{T<t+(d-1)\}] \\
&\quad = P[\{t-1<T<t\}] = e^{-\alpha(t-1)}e^{-\alpha t}
\end{aligned}$$



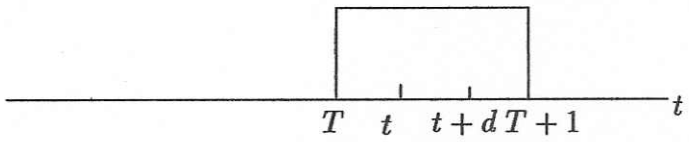

$$\begin{aligned}
&P[Y(t)=0, Y(t+d)=1] \\
&\quad = P[\{T>t\} \cap \{T<t+d<T+1\}] \\
&\quad = P[\{T>t\} \cap \{t+d-1<T<t+d\}] \\
&\quad = P[t+d-1<T<t+d] \\
&\quad = e^{-\alpha(t+d-1)} - e^{-\alpha(t+d)}
\end{aligned}$$




$$\begin{aligned}
&P[Y(t)=0, Y(t+d)=0] \\
&\quad = P[\{T+1<t\} \cup \{T>t+d\}] \\
&\qquad \cup \{\{t<T\} \cap \{T+1<t+d\}\}
\end{aligned}$$

$$\begin{aligned}
&P[Y(t)=0, Y(t+d)=0] \\
&\quad = P[T \le t-1] + P[T \ge t+d] + P[t \le T \le t+d-1] \\
&\quad = 1 - e^{-\alpha(t-1)} + e^{-\alpha(t+d)} \\
&\qquad + e^{-\alpha t} - e^{-\alpha(t+d-1)}
\end{aligned}$$

**Case 2:** $0 < d < 1$




$$\begin{aligned}
&P[Y(t) = 1, Y(t+d) = 1] \\
&\quad = P[\{T < t\} \cap \{t + d < T + 1\}] \\
&\quad = P[\{T < t\} \cap \{t + d - 1 < T\}] = P[t + d - 1 < T < t] \\
&\quad = e^{-\alpha(t+d-1)} - e^{-\alpha t}
\end{aligned}$$

$$\begin{aligned}
&P[Y(t) = 1, Y(t+d) = 0] \\
&\quad = P[\{t - 1 < T < t\} \cap \{T < t + (d-1)\}] \text{ as in Case 1} \\
&\quad = P[\{t - 1 < T < t + d - 1\}] \\
&\quad = e^{-\alpha(t-1)} - e^{-\alpha(t+d-1)}
\end{aligned}$$

$$\begin{aligned}
&P[Y(t) = 0, Y(t+d) = 1] \\
&\quad = P[\{T > t\} \cap \{t + d - 1 < T < t + d\}] \text{ as in Case 1} \\
&\quad = P[t < T < t + d] \\
&\quad = e^{-\alpha t} - e^{-\alpha(t+d)}
\end{aligned}$$

$$\begin{aligned}
&P[Y(t) = 0, Y(t+d) = 0] \\
&\quad = P[\{T + 1 < t\} \cup \{T > t + d\}] \\
&\quad = P[T < t - 1] + P[T > t + d] \\
&\quad = 1 - e^{-\alpha(t-1)} + e^{-\alpha(t+d)}
\end{aligned}$$

**c)**

$$\begin{aligned}
m_Y(t) &= 1 \cdot P[Y(t) = 1] + 0 P[Y(t) = 0] \\
&= e^{-\alpha(t-1)} - e^{-\alpha t}
\end{aligned}$$

$$\begin{aligned}\mathcal{E}[Y(t)Y(t+d)] &= 1 \cdot 1 \cdot P[Y(t)=1, Y(t+d)=1] \\ &= \begin{cases} e^{-\alpha(t+d-1)} - e^{-\alpha t} & 0 < d < 1 \\ 0 & \text{otherwise} \end{cases}\end{aligned}$$

$$\begin{aligned}C_Y(t,t+d) &= \mathcal{E}[Y(t)Y(t+d)] - m_Y(t)m_Y(t+d) \\ &= e^{-\alpha(t+d-1)} - e^{-\alpha t} \\ &\quad -(e^{-\alpha(t-1)} - e^{-\alpha t})(e^{-\alpha(t+d-1)} - e^{-\alpha(t+d)})\end{aligned}$$

9.9

$Z(t) = At^3 + B \qquad Z(t) = B \text{ if } t = 0$

**a)**

$$\begin{aligned}f_{Z(t)}(z)dz &= P[z < Z(t) \le z + dz] \text{ assume } t \neq 0 \\ &= P[z < At^3 + B \le z + dz] \\ &= \int_{-\infty}^{\infty} P[z < At^3 + b \le z + dz] f_B(b)db \\ &= \int_{-\infty}^{\infty} P\left[\frac{z-b}{t^3} < A \le \frac{z-b}{t^3} + \frac{dz}{t^3}\right] f_B(b)db \\ &= \int_{-\infty}^{\infty} f_A\left(\frac{z-b}{t^3}\right)\frac{dz}{|t^3|} f_B(b)db \\ \therefore f_{Z(t)}(z) &= \int_{-\infty}^{\infty} \frac{1}{|t^3|} f_A\left(\frac{z-b}{t^3}\right) f_B(b)db\end{aligned}$$

This is the convolution of the pdf of $At^3$ and $B$.

**b)**

$$\begin{aligned}\mathcal{E}[Z(t)] &= \mathcal{E}[At^3 + B] \\ &= \mathcal{E}[A]t^3 + \mathcal{E}[B]\end{aligned}$$

$$\mathcal{E}[Z(t_1)Z(t_2)] = \mathcal{E}[(At_1^3 + B)(At_2^3 + B)] = \mathcal{E}[A^2]t_1^3 t_2^3 + \mathcal{E}[AB](t_1^3 + t_2^3) + \mathcal{E}[B^2]$$

9.10

$$\begin{aligned}\mathcal{E}[|X_{t_2} - X_{t_1}|^2] &= \mathcal{E}[(X_{t_2} - X_{t_1})^2] \\ &= \mathcal{E}[X_{t_2}^2 - 2X_{t_2}X_{t_1} + X_{t_1}^2] \\ &= \mathcal{E}[X_{t_2}^2] - 2\mathcal{E}[X_{t_2}X_{t_2}] + \mathcal{E}[X_{t_1}^2] \\ &= R_X(t_2,t_2) - 2R_X(t_2,t_1) + R_X(t_1,t_1)\end{aligned}$$

9.11

~~6.10~~ **a)** $P[H(t)=1]=P[X(t)\geq 0]=P[\xi\cos 2\pi t\geq 0]=\frac{1}{2}=P[H(t)=-1]$

$$\begin{aligned}\mathcal{E}[H(t)] &= 1\cdot P[H(t)=1]+(-1)P[H(t)=-1]=0\\ C_H(t,t+\tau) &= \mathcal{E}[H(t)H(t+\tau)]\\ &= 1\cdot P[\underbrace{H(t)H(t+\tau)=1}_{\substack{H(t)\ \&\ H(t+\tau)\\ \text{same sign}}}]+(-1)P[\underbrace{H(t)H(t+\tau)=-1}_{\substack{H(t)\ \&\ H(t+\tau)\\ \text{opposite sign}}}]\end{aligned}$$

$$\begin{aligned}H(t)H(t+\tau) &= 1 \Leftrightarrow \cos 2\pi t \text{ and } \cos 2\pi(t+\tau) \text{ have same sign}\\ H(t)H(t+\tau) &= -1 \Leftrightarrow \cos 2\pi t \text{ and } \cos 2\pi(t+\tau) \text{ have different sign}\end{aligned}$$

$$\therefore C_H(t,t+\tau)=\begin{cases}1 & \text{for } t,\tau \text{ such that } \cos 2\pi t\cos 2\pi(t+\tau)=1\\ -1 & \text{for } t,\tau \text{ such that } \cos 2\pi t\cos 2\pi(t+\tau)=-1\end{cases}$$

**b)** $P[H(t)=1]=P[X(t)\geq 0]=P[\cos(\omega t+\Theta)\geq 0]=\frac{1}{2}=P[H(t)=-1]$

$$\begin{aligned}\mathcal{E}[H(t)] &= 1\left(\frac{1}{2}\right)+(-1)\frac{1}{2}=0\\ \mathcal{E}[H(t)H(t+\tau)] &= 1\cdot\underbrace{P[X(t)X(t+\tau)>0]}_{1-P[X(t)X(t+\tau)<0]}+(-1)P[X(t)X(t+\tau)<0]\\ &= 1-2P[X(t)X(t+\tau)<0]\end{aligned}$$

$$\begin{aligned}P[X(t)X(t+\tau)<0] &= P[\cos(\omega t+\Theta)\cos(\omega(t+\tau)+\Theta)<0]\\ &= \left[\frac{1}{2}\cos\omega\tau+\frac{1}{2}\cos(2\omega t+\omega\tau+2\Theta)<0\right]\\ &= P[\cos(2\omega t+\omega\tau+2\Theta)<\cos\omega\tau]\\ &= 1-\frac{\text{shaded region in figure}}{2\pi}\end{aligned}$$

**c)** $P[H(t)=1]=P[X(t)\geq 0]=1-F_{X(t)}(0^-)=1-P[H(t)=-1]$

$$\begin{aligned}\mathcal{E}[H(t)] &= 1\cdot P[H(t)=1]+(-1)P[H(t)=-1]\\ &= 1-F_{X(t)}(0^-)-F_{X(t)}(0^-)\\ &= 1-2F_{X(t)}(0^-)\end{aligned}$$

9.12

(a) $X(t)$ and $Y(t)$ independent implies $X(t)$ and $Y(t)$ uncorrelated

Since $\text{cov}(X(t_1), Y(t_2)) = E[X(t_1)Y(t_2)] - E[X(t_1)]E[Y(t_2)]$

$= E[X(t_1)]E[Y(t_2)] - E[X(t_1)]E[Y(t_2)]$

$= 0$

(d) If $X(t)$ and $Y(t)$ are uncorrelated then

$E[X(t_1)Y(t_2)] = E[X(t_1)]E[Y(t_2)]$

$X(t)$ and $Y(t)$ will be orthogonal if $X(t)$ and/or $Y(t)$ is zero mean.

(b) If $X(t)$ and $Y(t)$ are orthogonal, then

$E[X(t_1)Y(t_2)] = 0$, so

$\text{cov}(X(t_1), Y(t_2)) = E[X(t_1)]E[Y(t_2)]$

then $X(t)$ and $Y(t)$ will be uncorrelated if either $X(t)$ and/or $Y(t)$ are zero mean.

(c) Uncorrelated random variables are not necessarily independent and so uncorrelated random processes are not necessarily independent.

9.13

$$E[Z(t)] = E[2Xt - Y] = 2E[X]t - E[Y]$$
$$= 2tm_X - m_Y$$
$$\triangleq m_Z(t)$$

$$C_Z(t_1,t_2) = E[(2Xt_1 - Y)(2Xt_2 - Y)] - m_Z(t_1)m_Z(t_2)$$
$$= 4t_1t_2E[X^2] - 2(t_1+t_2)E[XY] + E[Y^2]$$
$$- 4t_1t_2m_X^2 + 2(t_1+t_2)m_Xm_Y - m_Y^2$$
$$= 4t_1t_2\sigma_X^2 - 2(t_1+t_2)\sigma_X\sigma_Y\rho_{XY} + \sigma_Y^2$$

$$\sigma_{Z(t)}^2 = C_Z(t,t) = 4t^2\sigma_X^2 - 4t\sigma_X\sigma_Y\rho_{XY} + \sigma_Y^2$$

$$f_{Z(t)}(z) = \frac{\exp\left\{-\dfrac{(z - 2tm_X + m_Y)^2}{2(4t^2\sigma_X - 4t\sigma_X\sigma_Y\rho_{XY} + \sigma_Y^2)}\right\}}{\sqrt{2\pi(4t^2\sigma_X - 4t\sigma_X\sigma_Y\rho_{XY} + \sigma_Y^2}}$$

9.14

a) $E[H(t)X(t)] = E[|X(t)|]$ since $H(t)X(t) = \begin{cases} X(t), & X(t) \geq 0 \\ -X(t), & X(t) < 0 \end{cases}$

$E[|X(t)|] = E[|\xi||\cos(2\pi t)|] = E[|\xi|\,|\cos(2\pi t)|] = |\cos 2\pi t|\,E[|\xi|] = \frac{1}{2}|\cos 2\pi t|$

My cos is cs, My sin is sin ← Note!

$E[H(t)X(t)] - E[H(t)]E[X(t)] = E[|X(t)|] - 0 \times E[X(t)] = E[|X(t)|] = \frac{1}{2}|\cos 2\pi t|$

Not Uncorrelated, Not Orthogonal

b)
Again: $E[H(t)X(t)] = E[|X(t)|]$

$E[H(t)] = +1\,P\{X(t) > 0\} - 1 \times P\{X(t) < 0\} = 0$

$C_V(X,H) = E[X(t)H(t)] - E[X(t)]E[H(t)] = E[|X(t)|]$

$E[|X(t)|] = E[|\cos(2\pi t + \xi)|] = \frac{1}{2\pi}\int_{-\pi}^{\pi} |\cos(2\pi t + \xi)|\,d\xi = \frac{2}{\pi}$

So $C_V(X,H) = \frac{2}{\pi}$

Not Uncorrelated, Not Orthogonal

9.15

$Y_n = X_n + g(n)$

a) $E[Y_n] = E[X_n] + g(n)$

$VAR[Y_n] = VAR[X_n + g(n)] = VAR[X_n]$

b) $F_{Y_n}(x) = P[Y_n < x] = P[(X_n + g(n)) < x] = P[X_n < x - g(n)] = F_{X_n}(x - g(n))$

$$F_{Y_n, Y_{n+1}}(x_1, x_2) = P[Y_n < x_1, Y_{n+1} < x_2] = P[X_n < x_1 - g(n), X_{n+1} < x_2 - g(n+1)]$$

$$= F_{X_n, X_{n+1}}(x_1 - g(n), x_2 - g(n+1))$$

c) $R_Y(n_1, n_2) = E[Y_{n_1} Y_{n_2}] = E[(X_{n_1} + g(n_1))(X_{n_2} + g(n_2))]$

$$= E[X_{n_1} X_{n_2}] + g(n_1) E[X_{n_2}] + g(n_2) E[X_{n_1}] + g(n_1) g(n_2)$$

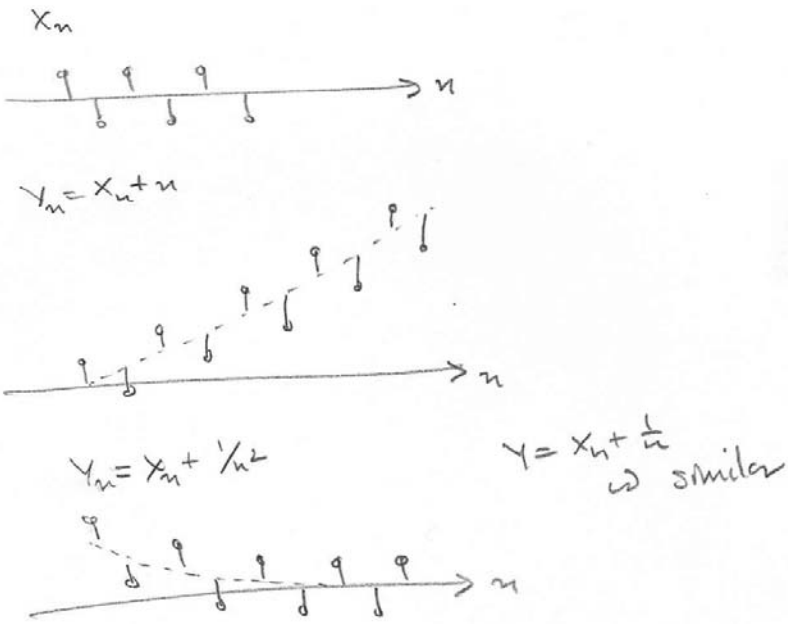
$$C_Y(n_1, n_2) = R_Y(n_1, n_2) - E[Y_{n_1}] E[Y_{n_2}] = E[X_{n_1} X_{n_2}] - E[X_{n_1}] E[X_{n_2}]$$

$$= C_X(n_1, n_2)$$

d) Based on $X_n$, $Y_n$ can be easily plotted

— continued —

9.15 (a)



9.16

a) $Y_n = c(n)X_n$

$$E[Y_n] = c(n)\,E[X_n]$$

$$VAR[Y_n] = VAR[c(n)X_n] = E\left[c^2(n)\left(E[X_n^2] - E^2[X_n]\right)\right] = c^2(n)\,VAR[X_n]$$

b)
$$F_{Y_n}(x) = P\{Y_n < x\} = P\{c(n)X_n < x\} = \begin{cases} P\{X_n < \frac{x}{c(n)}\} = F_{X_n}\left(\frac{x}{c(n)}\right) & \text{if } c(n) > 0 \\ P\{X_n > \frac{x}{c(n)}\} = 1 - F_{X_n}\left(\frac{x}{c(n)}\right) & \text{if } c(n) < 0 \\ u(x) & \text{if } c(n) = 0 \end{cases}$$

$$F_{Y_n, Y_{n+1}}(x_1, x_2) = P\{Y_n < x_1, Y_{n+1} < x_2\} = P\left\{c(n)X_n < x_1, c(n+1)X_{n+1} < x_2\right\}$$

$$= \begin{cases} F_{X_n, X_{n+1}}\left(\frac{x_1}{c(n)}, \frac{x_2}{c(n+1)}\right) & \text{if } c(n), c(n+1) > 0 \\ P\left\{X_n > \frac{x_1}{c(n)}, X_{n+1} < \frac{x_2}{c(n+1)}\right\} = F_{X_{n+1}}\left(\frac{x_2}{c(n+1)}\right) - F_{X_n, X_{n+1}}\left(\frac{x_1}{c(n)}, \frac{x_2}{c(n+1)}\right) & \text{if } c(n+1) > 0 \text{ \& } c(n) < 0 \\ P\left\{X_n < \frac{x_1}{c(n)}, X_{n+1} > \frac{x_2}{c(n+1)}\right\} = F_{X_n}\left(\frac{x_1}{c(n)}\right) - F_{X_n, X_{n+1}}\left(\frac{x_1}{c(n)}, \frac{x_2}{c(n+1)}\right) & \text{if } c(n+1) < 0, c(n) > 0 \\ P\left\{X_n > \frac{x_1}{c(n)}, X_{n+1} > \frac{x_2}{c(n+1)}\right\} = 1 - F_{X_n}\left(\frac{x_1}{c(n)}\right) - F_{X_{n+1}}\left(\frac{x_2}{c(n+2)}\right) + F_{X_n, X_{n+1}}\left(\frac{x_1}{c(n)}, \frac{x_2}{c(n+2)}\right) & \text{if } c(n) \text{ \& } c(n+1) < 0 \end{cases}$$

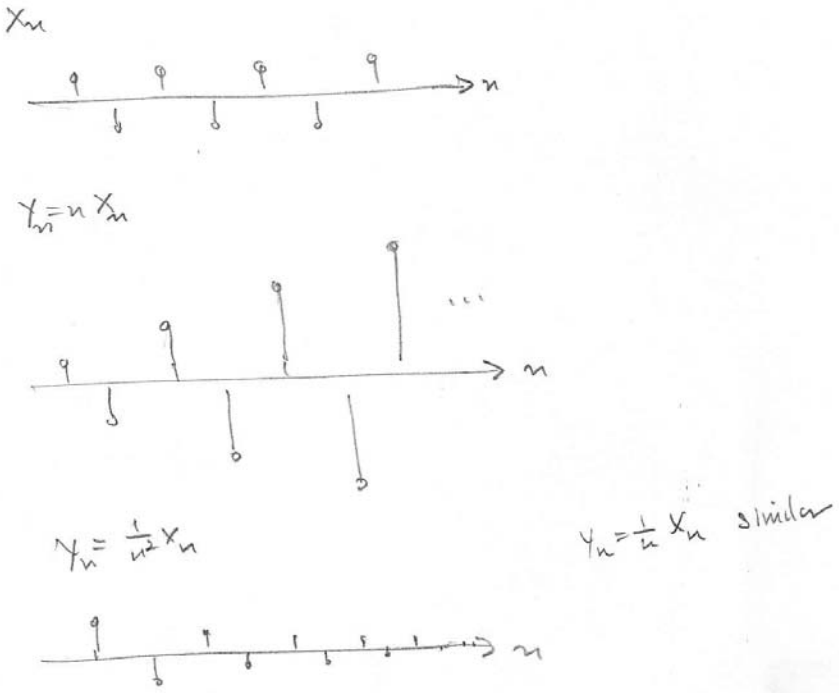
P9.16)

c) $R_Y(n_1,n_2) = E[Y_{n_1}Y_{n_2}] = c(n_1)c(n_2)\,E[X_{n_1}X_{n_2}] = c(n_1)c(n_2)R_X(n_1,n_2)$

$C_Y(n_1,n_2) = c(n_1)c(n_2)R_X(n_1,n_2) - c(n_1)c(n_2)E[X_{n_1}]E[X_{n_2}]$

$= c(n_1)c(n_2)C_X(n_1,n_2)$

d) Based on $X_n$, we can plot $Y_n$



9.17

a) $R_{X,Y}(n_1, n_2) = E[X_{n_1} Y_{n_2}] = E[X_{n_1}(X_{n_2} + g(n_2)] = E[X_{n_1} X_{n_2}] + g(n_2) E[X_{n_1}]$

$= R_X(n_1, n_2) + g(n_2) E[X_{n_1}]$

$C_{X,Y}(n_1, n_2) = R_{X,Y}(n_1, n_2) - E[Y_{n_1}] E[X_{n_2}] = R_X(n_1, n_2) + g(n_2) E[X_{n_1}]$

$- E[X_{n_1}] E[X_{n_2}] - g(n_2) E[X_{n_1}]$

$= R_X(n_1, n_2) - E[X_{n_1}] E[X_{n_2}] = C_X(n_1, n_2)$

b) $P\{X_n < x, Y_{n+1} < y\} = P\{X_n < x, X_{n+1} < y - g(n+1)\}$

$= F_{X_n, X_{n+1}}(x, y - g(n+1))$

---

c). $X_n$ and $Y_n$ are not independent since $Y_n = X_n + g(n)$.

$X_n$ and $Y_n$ are orthogonal if $R_{XY}(t_1, t_2) = 0$ <u>all</u> $t_1, t_2$

but $R_{XY}(n, n) = \underbrace{R_X(n, n)}_{\geq 0} + g(n) E[X_n]$

so $X_n$ and $Y_n$ are not orthogonal (except in very contrived cases).

Similarly $X_n$ and $Y_n$ are not uncorrelated.

9.18

a) $R_{X,Y}(n_1,n_2) = E[X_{n_1} Y_{n_2}] = c(n_2) R_X(n_1,n_2)$

$C_{X,Y}(n_1,n_2) = c(n_2) C_X(n_1,n_2)$

b) $$P[X_n < x, Y_{n+1} < y] = \begin{cases} F_{X_n,X_{n+1}}\left(x, \frac{y}{c(n+1)}\right) & \text{if } c(n+1) > 0 \\ F_{X_n}(x) - F_{X_n,X_{n+1}}\left(x, \frac{y}{c(n+1)}\right) & \text{if } c(n+1) < 0 \end{cases}$$

---

c). $X_n$ and $Y_n$ are orthogonal if

$R_{XY}(n_1,n_2) = 0$ for all $n_1, n_2$

$\Leftrightarrow$ $c(n_2) R_X(n_1,n_2) = 0$ all $n_1, n_2$

However in general $R_X(n,n) > 0$ and $c(n) > 0$ for at least some $n \Rightarrow X_n$ and $Y_n$ are not orthogonal

Similarly $X_n$ and $Y_n$ are not uncorrelated

To test independence consider

$$P[X_n < x, Y_n < y] = P[X_n < x, c(n) X_n < y] \quad \text{assume } c(n) > 0$$
$$= P\left[X_n < x, X_n < \frac{y}{c(n)}\right]$$
$$= P\left[X_n < \min\left(x, \frac{y}{c(n)}\right)\right]$$
$$\neq P[X_n < x] P[Y_n < y].$$

$X_n$ and $Y_n$ are not independent.

9.19 $U(t) = X(t) - Y(t)$ , $V(t) = X(t) + Y(t)$

a) $C_{UX}(t_1,t_2) = E[U(t_1)X(t_2)] - E[U(t_1)]E[X(t_2)]$

$= E[X(t_1)X(t_2) - Y(t_1)X(t_2)] - E[X(t_1)]E[X(t_2)] + E[Y(t_1)]E[X(t_2)]$

$= C_X(t_1,t_2) - C_{YX}(t_1,t_2) = C_X(t_1,t_2)$ (X & Y are ind.)

$C_{UY}(t_1,t_2) = E[U(t_1)Y(t_2)] - E[U(t_1)]E[Y(t_2)]$

$= E[X(t_1)Y(t_2) - Y(t_1)Y(t_2)] - E[X(t_1)]E[Y(t_2)] + E[Y(t_1)]E[Y(t_2)]$

$= C_{XY}(t_1,t_2) - C_Y(t_1,t_2) = -C_Y(t_1,t_2)$

$C_{UV}(t_1,t_2) = E[(X(t_1) - Y(t_1))(X(t_2)+Y(t_2))] - E[X(t_1)-Y(t_1)]E[X(t_2)+Y(t_2)]$

$= C_X(t_1,t_2) + C_{XY}(t_1,t_2) - C_{YX}(t_1,t_2) - C_Y(t_1,t_2)$

$= C_X(t_1,t_2) - C_Y(t_1,t_2)$

P 9.19)

b) for $f_{U(t_1)X(t_2)}(u,x)$ we define two aux. variables W & Z:

$$U(t_1) = X(t_1) - Y(t_1)$$
$$W(t_2) = X(t_2)$$
$$Z(t_1) = Y(t_1)$$

$$\Rightarrow \begin{bmatrix} U(t_1) \\ W(t_2) \\ Z(t_1) \end{bmatrix} = A \begin{bmatrix} X(t_1) \\ Y(t_1) \\ X(t_2) \end{bmatrix}, \quad A = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \quad \det(A) = -1$$

$$A^{-1} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$f_{U(t_1)W(t_2)Z(t_1)}(u,w,z) = f_{X(t_1)Y(t_1)X(t_2)}(u+z, z, w) = f_{Y(t_1)}(z)\, f_{X(t_1)X(t_2)}(u+z, w)$$

$$f_{U(t_1)X(t_2)}(u,x) = \int_{-\infty}^{+\infty} f_{U(t_1)W(t_2)Z(t_1)}(u,x,z)\,dz = \int_{-\infty}^{+\infty} f_{Y(t_1)}(z)\, f_{X(t_1)X(t_2)}(u+z,x)\,dz$$

for $f_{U(t_1)V(t_2)}(u,v)$ we need to define two aux. variables as well:

$$U(t_1) = X(t_1) - Y(t_1)$$
$$V(t_2) = X(t_2) + Y(t_2)$$
$$W(t_1) = Y(t_1)$$
$$Z(t_2) = Y(t_2)$$

, Trans. Matrix, $A = \begin{bmatrix} 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$, $A^{-1} = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$, $\det(A) = -1$

Therefore $f_{U(t_1)V(t_2)}(u,v) = \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} f_{UVWZ}(u,v,w,z)\,dw\,dz$

$$= \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} f_{X(t_1)Y(t_1)X(t_2)Y(t_2)}(u+w, w, v-z, z)\,dw\,dz$$

$$= \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} f_{X(t_1)X(t_2)}(u+w, v-z)\, f_{Y(t_1)Y(t_2)}(w,z)\,dw\,dz$$

9.20

a) $C_{UX}(n_1,n_2) = E[U(n_1)X(n_2)] - E[U(n_1)]E[X(n_2)]$

$= C_X(n_1,n_2) - C_{YX}(n_1,n_2) = C_X(n_1,n_2) = \begin{cases} \sigma_X^2 & n_1 = n_2 \\ 0 & n_1 \neq n_2 \end{cases}$

$C_{UY}(n_1,n_2) = E[U(n_1)Y(n_2)] - E[U(n_1)]E[Y(n_2)]$

$= C_{XY}(n_1,n_2) - C_Y(n_1,n_2) = -C_Y(n_1,n_2) = \begin{cases} -\sigma_Y^2 & n_1 = n_2 \\ 0 & n_1 \neq n_2 \end{cases}$

$C_{UV}(n_1,n_2) = E[U(n_1)V(n_2)] - E[U(n_1)]E[V(n_2)]$

$= C_X(n_1,n_2) + C_{XY}(n_1,n_2) - C_{YX}(n_1,n_2) - C_Y(n_1,n_2)$

$= C_X(n_1,n_2) - C_Y(n_1,n_2)$

since X & Y are different iid random processes:

$$C_{UV}(n_1,n_2) = \begin{cases} \sigma_X^2 - \sigma_Y^2 & n_1 = n_2 \\ 0 & n_1 \neq n_2 \end{cases}$$

b) $f_{U(t_1)X(t_2)}(u,x) = \int_{-\infty}^{+\infty} f_Y(z) f_X(u+z) f_X(x)\, dz =$

$= f_X(x) \cdot f_X(-u) * f_Y(u)$ , if $t_1 \neq t_2$

$f_{U(t_2)V(t_2)}(u,v) = \iint_{-\infty}^{+\infty} f_X(u+w) f_X(v-z) f_Y(w) f_Y(z)\, dw\, dz$

$= f_X(v) * f_Y(v) \cdot f_X(-v) * f_Y(u)$ if $t_1 \neq t_2$

if $t_1 = t_2$, $f_{U(t_1)X(t_1)}(u,x) = f_{XY}(x, x+v) = f_X(x) f_Y(x+u)$

$f_{U(t_1)V(t_2)}(u,v) = f_{XY}\left(\frac{1}{2}(u+v), \frac{1}{2}(v-u)\right) = f_X\left(\frac{1}{2}(u+v)\right) f_Y\left(\frac{1}{2}(v-u)\right)$

## 9.3 Sum Process, Binomial Counting Process, and Random Walk

9.21 ~~6.20~~ (a)

$$\begin{aligned} P[Y_n = 1] &= P[I_n \text{ is not erased } | I_n = 1]P[I_n = 1] \\ &= (1-\alpha)p \text{ where } I_n \text{ is Bernoulli process} \end{aligned}$$

The $Y_n$ are then a Bernoulli process with success probability

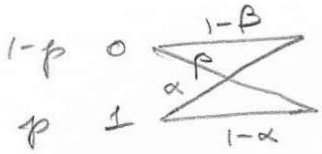
$$(1-\alpha)p \triangleq p'.$$

$S'_n$ is then the binomial counting process with

$$P[S'_n = k] = \binom{n}{k} p'^k (1-p')^{n-k}$$

$S'_n$ has independent and stationary increments.

(b)




$(1-p)(1-\beta) + p\alpha$

$p(1-p) + p(1-\alpha) \triangleq p''$

$S''_n$ is a Binomial counting process with $p''$

$S''_n$ has stationary and independent increments.

9.22

**6.21** a) Assume $n' > n$, $i \geq j$

$$\begin{aligned} P[S_n = j, S_{n'} = i] &= P[S_n = j, \overbrace{S_{n'-n} = i - j}^{\text{increment}}] \\ &= P[S_n = j]P[S_{n'-n} = i - j] \\ &\quad \text{by indep. increment property} \end{aligned}$$

In general

$$\begin{aligned} P[S_{n'} = i] &\neq P[S_{n'-n} = i - j] \\ \therefore P[S_n = j, S_{n'} = i] &\neq P[S_n = j]P[S_{n'} = i]. \end{aligned}$$

b)

$$\begin{aligned} P[S_{n_2} = j | S_{n_1} = i] &= \frac{P[S_{n_2} = j, S_{n_1} = i]}{P[S_{n_1} = i]} \\ &= \frac{P[\overbrace{S_{n_2} - S_{n_1} = j - i}^{\text{increment}}]P[S_{n_1} = i]}{P[S_{n_1} = i]} \\ &= P[S_{n_2} - S_{n_1} = j - i] = \binom{n_2 - n_1}{j - 0} p^{j-i}(1-p)^{n_2 - n_1 - j + i} \end{aligned}$$

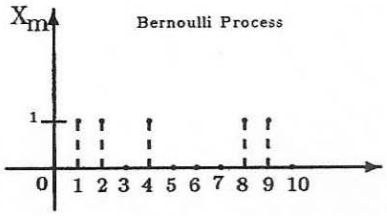
$$\begin{aligned} P[S_{n_2} | S_{n_1} = i, S_{n_0} = k] &= \frac{P[S_{n_2} = j, S_{n_1} = i, S_{n_0} = k]}{P[S_{n_1} = i, S_{n_0} = k]} \\ &= \frac{P[S_{n_0} = k, S_{n_1} - S_{n_0} = i - k, S_{n_2} - S_{n_1} = j - i]}{P[S_{n_0} = k, S_{n_1} - S_{n_0} = i]} \\ &= \frac{P[S_{n_0} = k]P[S_{n_1} - S_{n_0} = i - k]P[S_{n_2} - S_{n_1} = j - i]}{P[S_{n_0} = k]P[S_{n_1} - S_{n_0} = i]} \\ &= P[S_{n_2} - S_{n_1} = j - i] \\ &= P[S_{n_2} = j | S_{n_1} = i]. \end{aligned}$$

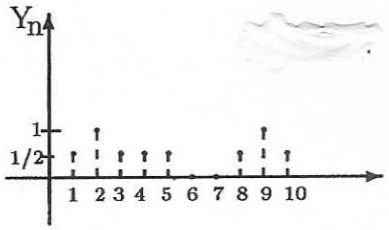
9.23

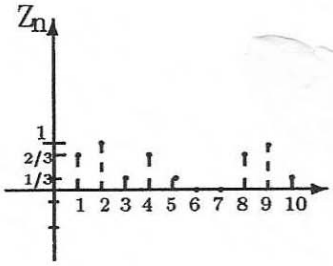
$$\begin{aligned} S_n = 0 &\Leftrightarrow \#\ 1\text{'s} = \#\ 0\text{'s} \\ &\Leftrightarrow n \text{ is even and } \#\ 1\text{'s} = \#\ 0\text{'s} = \frac{n}{2} \end{aligned}$$

$$\therefore P[X_n = 0] = \begin{cases} \binom{n}{\frac{n}{2}} p^{\frac{n}{2}}(1-p)^{\frac{n}{2}} & n \text{ even} \\ 0 & n \text{ odd} \end{cases}$$

9.24




$$
\begin{aligned}
\mathcal{E}[Y_n] &= \frac{1}{2}\mathcal{E}[X_n] + \frac{1}{2}\mathcal{E}[X_{n-1}] = \frac{1}{2}p + \frac{1}{2}p = p \\
\mathcal{E}[Y_n^2] &= \mathcal{E}\left[\frac{1}{4}X_n^2 + \frac{2}{4}X_nX_{n-1} + \frac{1}{4}X_{n-1}^2\right] \\
&= \frac{1}{4}p\underbrace{\mathcal{E}[X_n^2]}_{p} + 2\underbrace{\mathcal{E}[X_n]}_{p}\underbrace{\mathcal{E}[X_{n-1}]}_{p} + \underbrace{\mathcal{E}[X_{n-1}^2]}_{p} \\
&= \frac{1}{2}p(1+p) \\
\mathcal{E}[Y_nY_{n+1}] &= \frac{1}{4}\mathcal{E}[X_nX_{n+1} + X_n^2 + X_{n-1}X_{n+1} + X_{n-1}X_n] \\
&= \frac{1}{4}[p + 3p^2] \\
\mathcal{E}[Y_nY_{n+1}] &= \mathcal{E}\left[\frac{1}{4}(X_n + X_{n-1})(X_{n+1} + X_n)\right] = \frac{1}{4} + \frac{3}{4}(2p-1)^2
\end{aligned}
$$

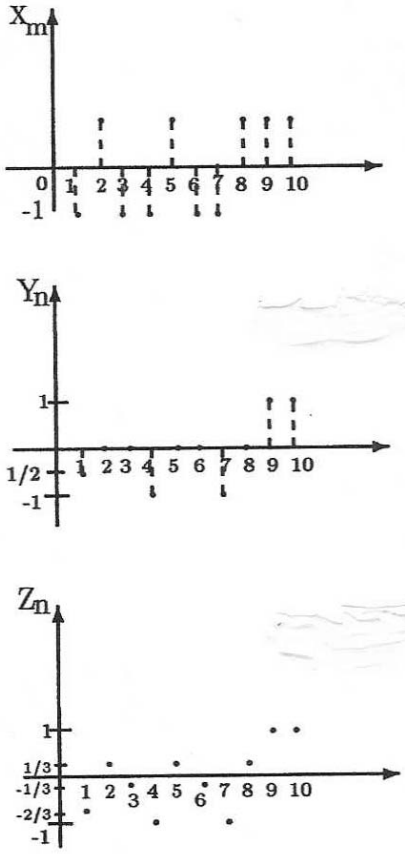
For $k > 1$ $\mathcal{E}[Y_n Y_{n+k}] = \mathcal{E}[Y_n]\mathcal{E}[Y_{n+k}] = p^2$

$$\therefore C_Y(n, n+k) = \begin{cases} \frac{p}{2} - \frac{p^2}{2} & k = 0 \\ \frac{p}{4} - \frac{p^2}{4} & k = 1 \\ 0 & k > 1 \end{cases}$$

$$\begin{aligned} \mathcal{E}[Z_n] &= \frac{2}{3}\mathcal{E}[X_n] + \frac{1}{3}\mathcal{E}[X_{n-1}] = p \\ \mathcal{E}[Z_n^2] &= \frac{1}{9}\mathcal{E}[(4X_n^2 + 4X_n X_{n-1} + X_{n-1}^2)] = \frac{5}{9}p + \frac{4}{9}p^2 \\ \mathcal{E}[Z_n Z_{n+1}] &= \frac{1}{9}\mathcal{E}[4X_n X_{n+1} + 2X_n^2 + 2X_{n+1}X_{n-1} + X_n X_{n-1}] = \frac{7}{9}p^2 + \frac{2}{9}p \\ \mathcal{E}[Z_n Z_{n+k}] &= \mathcal{E}[Z_n]\mathcal{E}[Z_{n+k}] = p^2 \text{ for } k > 1 \end{aligned}$$

$$\therefore C_Z(n, n+k) = \begin{cases} \frac{5}{9}p - \frac{5}{9}p^2 & k = 0 \\ \frac{2}{4}p - \frac{2}{9}p^2 & k = 1 \\ 0 & k > 1 \end{cases}$$

Random Step Process



$$\begin{aligned}
\mathcal{E}[Y_n] &= \frac{1}{2}\mathcal{E}[X_n] + \frac{1}{2}\mathcal{E}[X_{n-1}] = 2p-1 \\
\mathcal{E}[Z_n] &= \frac{2}{3}\mathcal{E}[X_n] + \frac{1}{3}\mathcal{E}[X_{n-1}] = 2p-1 \\
\mathcal{E}[Y_n^2] &= \mathcal{E}\left[\frac{1}{4}(X_n^2 + 2X_nX_{n-1} + X_{n-1}^2)\right] \\
&= \frac{1}{4}\{1 + 2(2p-1)^2 + 1\} = \frac{1}{2} + \frac{(2p-1)^2}{2} \\
\mathcal{E}[Z_n^2] &= \frac{1}{9}\mathcal{E}[4X_n^2 + 4X_nX_{n-1} + X_{n-1}^2] \\
&= \frac{1}{9}[5 + 4(2p-1)^2] = \frac{5}{9} + \frac{4}{9}(2p-1)^2
\end{aligned}$$

For $k > 1$

$$\begin{aligned}
\mathcal{E}[Y_nY_{n+k}] &= \mathcal{E}[Y_n]\mathcal{E}[Y_{n+k}] = (2p-1)^2 \\
\mathcal{E}[Z_nZ_{n+1}] &= \mathcal{E}\left[\frac{1}{9}(2X_n + X_{n-1})(2X_{n+1} + X_n)\right] = \frac{2}{9} + \frac{7}{9}(2p-1)^2
\end{aligned}$$

For $k > 1$

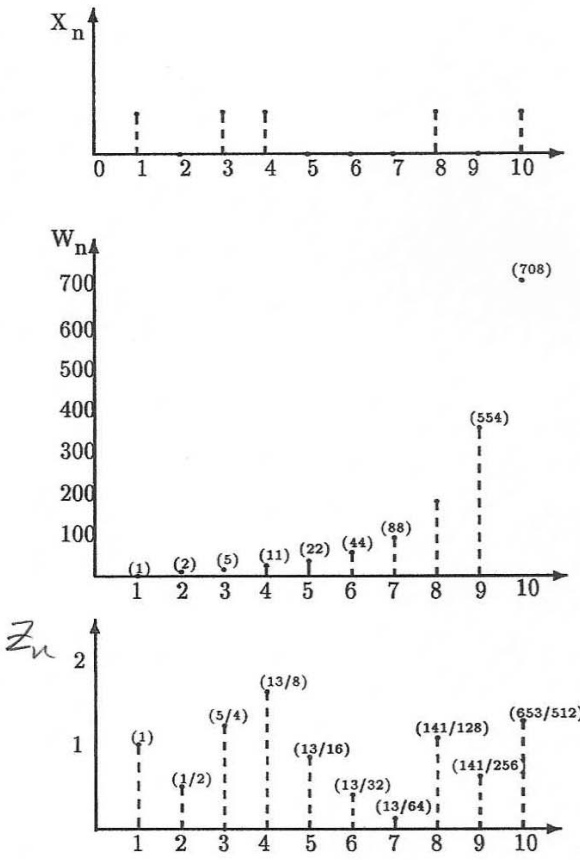
$$\mathcal{E}[Z_nZ_{n+1}] = \mathcal{E}[Z_n]\mathcal{E}[Z_{n+1}] = (2p-1)^2$$

$$\therefore C_Y(n, n+k) = \begin{cases} \frac{1}{2} - \frac{1}{2}(2p-1)^2 & k = 0 \\ \frac{1}{4} - \frac{1}{4}(2p-1)^2 & k = 1 \\ 0 & k > 1 \end{cases}$$

$$C_Z(n, n+k) = \begin{cases} \frac{5}{9} - \frac{5}{9}(2p-1)^2 & k = 0 \\ \frac{2}{9} - \frac{2}{9}(2p-1)^2 & k = 1 \\ 0 & k > 1 \end{cases}$$

In all cases, the sample functions are close to the mean of the processes.

9.25 a)




$(Z_n = \frac{3}{4}Z_{n-1} + \frac{1}{4}X_n)$ in this graph

$W_n$ is exponentially increasing without bound as $n \to \infty$ and has meaningless sample mean.

$Z_n$ is exponentially decreasing unless $X_n$ is 1 and hence, the sample mean is about twice of the sample mean of $X_n$.

b)
$$\begin{aligned} W_n &= 2W_{n-1} + X_n \quad n > 1 \\ &= 2(W_{n-2} + S_{n-1}) + X_n \\ &= X_n + 2X_{n-1} + 4X_{n-2} + \ldots + 2^{n-1}X_1 \\ \mathcal{E}[W_n] &= \mathcal{E}[X]\{1 + 2 + \ldots + 2^{n-1}\} = \mathcal{E}[X]\frac{1-2^n}{1-2} = (2^n - 1)\mathcal{E}[X] \\ Z_n &= \frac{3}{4}Z_{n-1} + X_n = \frac{3}{4}\left(\frac{3}{4}|Z_{n-2} + X_{n-1}\right) + X_n \\ &= X_n + \frac{3}{4}X_{n-1} + \ldots + \left(\frac{3}{4}\right)^{n-1}X_1 \\ \mathcal{E}[Z_n] &= \mathcal{E}[X]\left\{1 + \frac{3}{4} + \ldots + \left(\frac{3}{4}\right)^n\right\} = \mathcal{E}[X]\frac{1-\left(\frac{3}{4}\right)^{n-1}}{1-\frac{3}{4}} \end{aligned}$$

c) Since $W_n - W_{n-1} = W_{n-1} + X_n$, $W_n$ does not have independent increments.
Since $Z_n - Z_{n-1} = X_n - \frac{1}{2}Z_{n-1}$, $Z_n$ does not have independent increments.

9.26

**6.25** a) $\mathcal{E}[M_n] = \frac{1}{n}\mathcal{E}[X_1 + X_2 + ... + X_n] = \frac{n\mathcal{E}[X]}{n} = \mathcal{E}[X]$

$$\begin{aligned} C_M(n,k) &= \mathcal{E}[(M_n - \mathcal{E}(X))(M_k - \mathcal{E}(X))] \\ &= \mathcal{E}\left[\frac{1}{n}[S_n - n\mathcal{E}(X)]\frac{1}{k}[S_k - k\mathcal{E}[X]]\right] \\ &= \frac{1}{nk}\mathcal{E}[(S_n - \mathcal{E}[S_n])(S_k - \mathcal{E}[S_k])] \\ &= \frac{1}{nk}C_S(n,k) = \frac{1}{nk}\min(n,k)\sigma_X^2 \\ VAR(M_n) &= C_M(n,n) = \frac{1}{n}\sigma_X^2 \end{aligned}$$

b) Since $M_{n+1} - M_n = \frac{1}{n+1}X_n - \frac{1}{n+1}M_n$, $M_n$ does not have indep. increments.

9.27

$Y_n$ and $Z_n$ are Gaussian random processes with mean

$$\begin{aligned} \mathcal{E}[Y_n] &= \frac{1}{2}\mathcal{E}[X_n] + \frac{1}{2}\mathcal{E}[X_{n-1}] = 0 \\ \mathcal{E}[Z_n] &= \frac{2}{3}\mathcal{E}[X_n] + \frac{1}{3}\mathcal{E}[X_{n-1}] = 0 \end{aligned}$$

and variance

$$\begin{aligned} \mathcal{E}[Y_n^2] &= \mathcal{E}\left[\frac{1}{4}(X_n + X_{n-1}^2)\right] = \frac{1}{4}(1+1) = \frac{1}{2} \\ \mathcal{E}[Z_n^2] &= \mathcal{E}\left[\left(\frac{2}{3}X_n + \frac{1}{3}X_{n-1}\right)^2\right] = \frac{5}{9} \end{aligned}$$

$$\therefore f_{Y_n}(y) = \frac{e^{-y^2}}{\sqrt{\pi}} \qquad f_{Z_n}(z) = \frac{3}{\sqrt{10\pi}}e^{-9z^2/10}$$

9.28

6.27 a) $\Phi_{S_n}(\omega) = \Phi_X(\omega)^n = (e^{-\alpha|\omega|})^n = e^{-n\alpha|\omega|}$

$$\Rightarrow f_{S_n}(x) = \frac{n\alpha/\pi}{x^2 + n^2\alpha^2}$$

b) Since $S_n$ has independent and stat. increments

$$\begin{aligned} f_{S_n,S_{n+k}}(y_1, y_2) &= f_{S_n}(y_1) f_{S_{n+k-n}}(y_2 - y_1) \\ &= \frac{n\alpha/\pi}{(y_1^2 + n^2\alpha^2)} \frac{k\alpha/\pi}{((y_2 - y_1)^2 + k^2\alpha^2)} \end{aligned}$$

9.29

6.28 a) $G_{S_n}(z) = G_X(z)^n = (e^{\alpha(z-1)})^n = e^{n\alpha(z-1)}$

$$\Rightarrow P[S_n = k] = \frac{(n\alpha)^k}{k!} e^{-n\alpha}$$

b) $P[S_n = i, S_{n+k} = j] = P[S_n = i] P[S_{n+k-n} = j - i]$

$$= \frac{(n\alpha)^i}{i!} e^{-n\alpha} \frac{(k\alpha)^{j-1}}{(j-i)!} e^{-k\alpha} \quad \text{for } j \geq i,\ k > 0$$

9.30

6.29 a)

$$\begin{aligned} \Phi_{M_n}(\omega) &= \mathcal{E}[e^{j\omega(X_1 + \ldots + S_n)/n}] = \Phi_X^n\left(\frac{\omega}{n}\right) = e^{-n\left(\frac{\omega}{n}\right)^2/2} \\ &= e^{-\omega^2/2n} \end{aligned}$$

$\Rightarrow M_n$ is Gaussian with mean zero and variance $\frac{1}{n}$

$$F_{M_n}(x) = \sqrt{\frac{n}{2\pi}} e^{-nx^2/2}$$

b) $M_n$ and $M_{n+k}$ are related to $S_n$ and $S_{n+k}$ by

$$\begin{aligned} M_n &= \frac{1}{n} S_n \\ M_{n+k} &= \frac{1}{n+k} S_{n+k} \\ J(S_n, S_{n+k}) &= \begin{vmatrix} \frac{1}{n} & 0 \\ 0 & \frac{1}{n+k} \end{vmatrix} = \frac{1}{n(n+k)} \end{aligned}$$

$$\Rightarrow f_{M_n M_{n+k}}(x, y) = n(n+k) f_{S_n S_{n+k}}(nx, (n+k)y)$$

$$\begin{aligned} f_{M_n M_{n+k}}(x, y) &= f_{S_n}(nx) f_{S_k}((n+k)y - nx)(n)(n+k) \\ &= n(n+k) \frac{e^{-n^2x^2/2n}}{\sqrt{2\pi n}} \frac{e^{-[(n+k)y - nx]^2/2k}}{\sqrt{2\pi k}} \end{aligned}$$

since $S_n$ is Gaussian with mean zero and variance $n$.

**9.31** $X_n = \frac{1}{2}(Y_n + Y_{n-1})$ $Y_n$ iid.

$$E[X_n] = \frac{1}{2}[E[Y_n] + E[Y_{n-1}]] = E[Y] \triangleq m$$

$$\text{cov}(X_n, X_{n+k}) = E[(X_n - m)(X_{n+k} - m)] = E[X_n X_{n+k}] - m^2$$

$$= \begin{cases} \frac{1}{4}E[(Y_j + Y_{j-1})^2] - m^2 = \frac{1}{2}E[Y^2] - \frac{1}{2}m^2 = \frac{1}{2}\text{VAR}(Y) & \text{for } k=0 \\ \frac{1}{4}E[(Y_j + Y_{j-1})(Y_{j-1} + Y_j)] & \\ \quad = \frac{1}{4}E[Y^2] + \frac{3}{4}m^2 - m^2 = \frac{1}{4}\text{VAR}(Y) & k = \pm 1 \\ \frac{1}{4}E[(Y_j + Y_{j-1})]E[(Y_{j+k} + Y_{j+k-1})] = 0 & \text{otherwise} \end{cases}$$

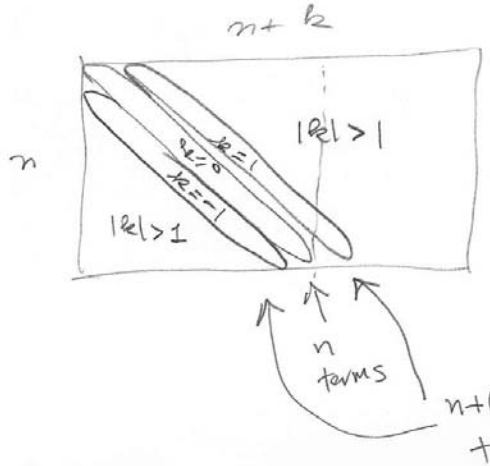
$$M_n = \frac{1}{n}\sum_{j=1}^{n} X_j \qquad E[M_n] = \frac{1}{n}\sum_{j=1}^{n} E[X_j] = m$$

for $k > 1$

$$\text{cov}(M_n, M_{n+k}) = E[(M_n - m)(M_{n+k} - m)]$$

$$= E\left[\frac{1}{n}\sum_{j=1}^{n}(X_j - m)\frac{1}{n+k}\sum_{l=1}^{n+k}(X_l - m)\right]$$

$$= \frac{1}{n(n+k)}\sum_{j=1}^{n}\sum_{l=1}^{n+k}\underbrace{E[(X_j - m)(X_l - m)]}_{\text{cov}(X_j, X_l)}$$




$$= \frac{1}{n(n+k)}\left\{n\frac{1}{2}\sigma_Y^2 + (2n-1)\frac{\sigma_Y^2}{4}\right\}$$

$$= \frac{4\sigma_Y^2(4n-1)}{n(n+k)} \qquad k \geq 1$$

$$\text{VAR}(M_n) = \frac{4\sigma_Y^2(4n-2)}{n(n+k)}$$

$\to 0$ as $n \to \infty$.

9.32

$$X_n = \alpha X_{n-1} + Y_n \qquad \alpha = 3/4$$

$$= Y_n + \alpha(\alpha X_{n-2} + Y_{n-1})$$

$$= Y_n + \alpha^2 Y_{n-1} + \alpha^2 Y_{n-2} + \ldots + \alpha^{n-1} Y_0$$

To simplify derivation assume process started at time $-\infty$

Then,

$$X_n = Y_n + \alpha^2 Y_{n-1} + \alpha^2 Y_{n-1} + \ldots$$

$$= \sum_{l=0}^{\infty} \alpha^l Y_{n-l}$$

$$E[X_n] = \sum_{l=0}^{\infty} \alpha^l \underbrace{E[Y_{n-l}]}_{m_Y} = m_Y \frac{1}{1-\alpha} \triangleq m_X$$

$$COV(X_n, X_{n+k}) = E[(X_n - m_X)(X_{n+k} - m_X)]$$

$$= E\left[\sum_{l=0}^{\infty} \alpha^l Y_{n-l} \sum_{l'=0}^{\infty} \alpha^{l'} Y_{n-l'+k}\right] - m_X^2$$

$$= \sum_{l=0}^{\infty}\sum_{l'=0}^{\infty} \alpha^{l+l'} \underbrace{E[Y_{n-l} Y_{n-l'+k}]}_{E[Y^2]\,\delta_{l-l'-k}} - m_X^2$$

$$= \alpha^k \sum_{l=0}^{\infty} \alpha^{2l} E[Y^2] + \sum_{\substack{l=0 \\ l \neq l'+k}}^{\infty}\sum_{l'=0}^{\infty} \alpha^{l+l'} m_Y^2 - m_X^2$$

$$= \frac{\alpha^k}{1-\alpha^2} E[Y^2] + \left(\left(\frac{1}{1-\alpha}\right)^2 - \frac{\alpha^k}{1-\alpha^2}\right) m_Y^2 - \underbrace{m_X^2}_{\frac{m_Y^2}{(1-\alpha)^2}}$$

$$= \frac{\alpha^k}{1-\alpha^2} \sigma_Y^2$$

9.32 — Now consider $M_n$ —

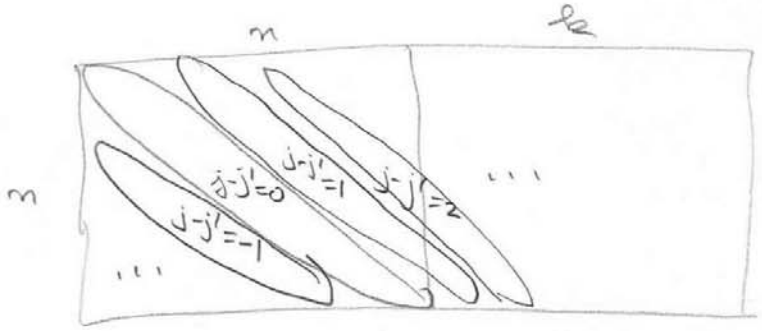
$$E[M_n] = \frac{1}{n}\sum_{j=1}^{n} E[X_j] = m_X$$

$$\text{cov}(M_n, M_{n+k}) = E\left[(M_n - m_X)(M_{n+k} - m_X)\right]$$

$$= \frac{1}{n(n+k)} E\left[\sum_{j=1}^{n}(X_j - m_X)\sum_{j'=1}^{n+k}(X_{j'} - m_X)\right]$$

$$= \frac{1}{n(n+k)} \sum_{j=1}^{n}\sum_{j'=1}^{n+k} \text{cov}(X_j, X_{j'})$$

$$\text{cov}(X_j, X_{j'}) = \frac{\alpha^{|j-j'|}}{1-\alpha^2}\sigma_Y^2$$




$$\text{cov}(M_n, M_{n+k}) = \frac{\sigma_Y^2/(1-\alpha^2)}{n(n+k)}\left(n\cdot\alpha^0 + (n+(n-1))\alpha^1 + (n+(n-2))\alpha^2 + \cdots\right)$$

$$\text{VAR}(M_n) = \frac{\sigma_Y^2/(1-\alpha^2)}{n^2}\sum_{k=-(n-1)}^{n-1}(n-|k|)\alpha^{|k|}$$

$$= \frac{\sigma_Y^2/(1-\alpha^2)}{n}\sum_{k=-(n-1)}^{n-1}\left(1-\frac{|k|}{n}\right)\alpha^{|k|}$$

$$\leq \frac{\sigma_Y^2/(1-\alpha^2)}{n}\cdot\frac{1}{1-\alpha} \to 0 \text{ as } n\to\infty.$$

9.33

~~6.30~~ a)

$$\begin{aligned}\underline{S}(n) &= (S_0(n), S_1(n), S_2(n)) \\ &= (X_0(1) + \ldots + S_0(n), S_1(1) + \ldots + X_1(n), X_2(n) + \ldots + X_2(n))\end{aligned}$$

$\Rightarrow S_i(n) =$ number of occurences of $i$ in $n$ trials

$S_0(n) + S_1(n) + S_2(n) = n$

$$\Rightarrow P[S_0(n) = i, S_1(n) = j, S_2(n) = n - j - i] = \frac{n!}{i!j!(n-i-j)!} p_0^i p_1^j p_2^{n-i-j}$$

$\Rightarrow \underline{S}(n)$ Has a multinomial dist.

b)
$$\left.\begin{aligned}\underline{S}_{n_1} - \underline{S}_{n_0} &= \underline{X}_{n_0+1} + \ldots + \underline{X}_{n_1} \\ \underline{S}_{n_3} - \underline{S}_{n_2} &= \underline{X}_{n_2+1} + \ldots + \underline{X}_{n_3}\end{aligned}\right\}$$

if $n_2 > n_1$ then $\underline{S}_{n_1} - \underline{S}_{n_0}$ and $\underline{S}_{n_3} - \underline{S}_{n_2}$ depends on different sets of $X$'s and hence they are independent

$$\begin{aligned}P[\underline{S}(n) = \underline{S}, \underline{S}(n+k) = \underline{S}'] &= P[\underline{S}(n) = \underline{S}] P[\underline{S}(k) = \underline{S}' - \underline{S}] \\ &= \frac{n!}{s_0! s_1! s_2!} \frac{k!}{(s_0' - s_0)!(s_1' - s_1)!(s_2' - s_2)!} p_0^{s_0'} p_1^{s_1'} p_2^{s_2'}\end{aligned}$$

c) $S_j(n) = X_j(1) + \ldots + X_j(n)$ where $X_j(i)$ are Bernoulli RV's

$\Rightarrow S_j(n)$ is a binomial counting process.

## 9.4 Poisson and Associated Random Processes

9.34

$\lambda = 10$ queries/minute

$T = 20\ \text{sec} = 1/3$ minute

$$P[N(\tfrac{1}{3}) = 0] = \frac{(10/3)^0}{0!} e^{-10/3} = 0.0356$$

9.35

$$P[N_1(t) = k \mid N(t) = n] = \binom{n}{k} p^k (1-p)^{n-k}$$

items dispensed

$$P[N_1(t) = k] = \sum_{l=k}^{\infty} \binom{l}{k} p^k (1-p)^{l-k} \frac{(\lambda t)^l}{l!} e^{-\lambda t}$$

$$P[N_1(t) = k] = \frac{e^{-\lambda t} p^k (\lambda t)^k}{k!} \sum_{l=k}^{\infty} \frac{((1-p)\lambda t)^{l-k}}{(l-k)!}$$

$$\sum_{l'=0}^{\infty} \frac{((1-p)\lambda t)^{l'}}{l'!} = e^{(1-p)\lambda t}$$

$$= \frac{(\lambda t p)^k}{k!} e^{-\lambda t p}$$

9.36

(a) $$P[N(t) = 0] = \frac{(\lambda t)^0}{0!} e^{-\lambda t} = e^{-\lambda t}$$

(b) $$P[N(t) > 2] = 1 - P[N(t) \le 2]$$

$$= 1 - e^{-\lambda t}\left(1 + \lambda t + \frac{(\lambda t)^2}{2}\right)$$

9.37

a)
$$P\{T_1<T_2\} = \int_0^\infty P[T_2>t \mid T_1=t]\, f_1(t)\,dt = \int_0^\infty e^{-\lambda_2 t}\,\lambda_1 e^{-\lambda_1 t}\,dt = \frac{\lambda_1}{\lambda_1+\lambda_2} = \frac{2}{3}$$

b) $f_1(t) = \lambda_1 e^{-\lambda_1 t}$, $f_2(t) = \lambda_2 e^{-\lambda_2 t}$, $F_1(t) = 1-e^{-\lambda_1 t}$, $F_2(t) = 1-e^{-\lambda_2 t}$

$$f_{min}(t) = f_1(t) + f_2(t) - f_1(t)F_2(t) - F_1(t)f_2(t)$$
$$= \lambda_1 e^{-\lambda_1 t} + \lambda_2 e^{-\lambda_2 t} - \lambda_1 e^{-\lambda_1 t} - \lambda_2 e^{-\lambda_2 t} + \lambda_1 e^{-(\lambda_1+\lambda_2)t} + \lambda_2 e^{-(\lambda_1+\lambda_2)t}$$
$$= (\lambda_1+\lambda_2)\, e^{-(\lambda_1+\lambda_2)t} = 3e^{-3t}$$

---

$$1-F_{min}(t) = P\{\min(T_1,T_2)>t\} = P[T_1>t, T_2>t] = P[T_1>t]P[T_2>t] = e^{-\lambda_1 t}\cdot e^{-\lambda_2 t}$$
$$= e^{-(\lambda_1+\lambda_2)t} = e^{-3t}$$

c) $N(t)$ is a poisson random variable with rate $(\lambda_1+\lambda_2)t$

Hence, $P\{N(t)=k\} = e^{-(\lambda_1+\lambda_2)t}\,\frac{(\lambda_1+\lambda_2)^k t^k}{k!} = e^{-3t}\,\frac{(3t)^k}{k!}$

Note that sum of $n$ independent poisson RVs is a poisson RV (P7.11)

d) at time $t$, $N_i(t)$ is a Poisson RV with rate $\lambda_i t$.
So, if we define $N(t) = \sum_{i=1}^{k} N_i(t)$, $N(t)$ is sum of $k$ independent random variable with rat $\sum_{i=1}^{k} \lambda_i t$. if we define $\lambda := \sum \lambda_i(t)$

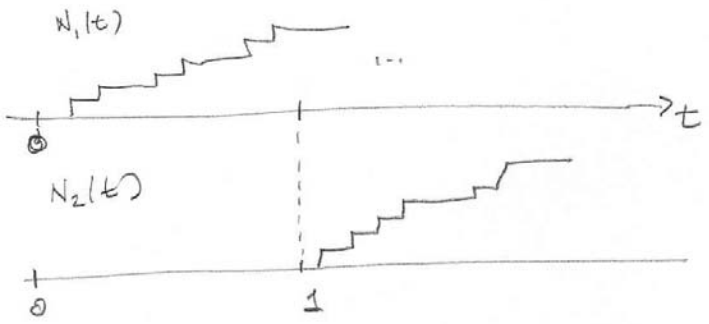
we have:
$$P\{N(t)=m\} = e^{-\lambda t}\,\frac{(\lambda t)^m}{m!}$$

it can be shown that $N(t)$ is a Poisson process with rate $\lambda$

9.38

$$\begin{aligned}
P[N(t-d)=j|N(t)=k] &= \frac{P[N(t-d)=j, N(t)=k]}{P[N(t)=k]} \\
&= \frac{P[N(t-d)=j]P[N(t)-N(t-d)=k-j]}{P[N(t)=k]} \\
&= \frac{\frac{\lambda^j(t-d)^j}{j!}e^{-\lambda(t-d)}\frac{\lambda^{k-j}d^{k-j}}{(k-j)!}e^{-\lambda d}}{\frac{\lambda^k t^k}{k!}e^{-\lambda t}} \\
&= \binom{k}{j}\left(\frac{t-d}{t}\right)^j\left(\frac{d}{t}\right)^{k-j} \quad \text{binomial}
\end{aligned}$$

9.39




$$N(t) = \begin{cases} N_1(t) & 0 \le t < 1 \\ N_1(t) + N_2(t) & 1 \le t \end{cases} \qquad m(t) = \begin{cases} \lambda_1 t & 0 \le t < 1 \\ (\lambda_1 + \lambda_2)t & 1 \le t \end{cases}$$

Since $N_1(t)$ and $N_2(t)$ are independent random processes

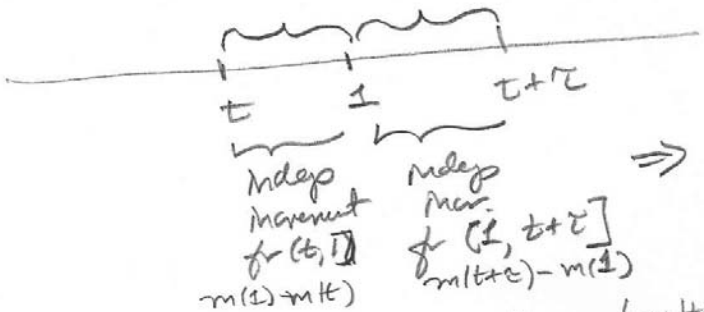
$N_1(t) + N_2(t)$ has a Poisson pmf.

$$P[N(t) = k] = P[N_1(t) + N_2(t) = k] = \frac{((\lambda_1+\lambda_2)t)^k}{k!} e^{-(\lambda_1+\lambda_2)t}$$

for $t \ge 1$.

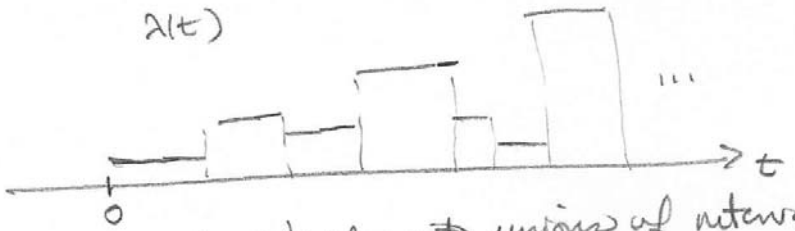
$N(t)$ has indep increments for intervals in $t < 1$, and for $t > 1$

$N(t)$ also has indep increments for intervals that span $t = 1$:




$$\therefore \quad P[N(t+\tau) - N(t) = k] = \frac{(m(t+\tau) - m(t))^k}{k!} e^{-(m(t+\tau)-m(t))}$$

(b) The above 2-interval arrival rate process is generalized to a piecewise constant process




By decomposing intervals into unions of intervals where $\lambda(t)$ is constant, we can show that pmf in part (a) holds.

For the interval that begins at $t=0$, we have.

$$P[N(t)=k] = \frac{m(t)^k}{k!} e^{-m(t)} \quad \text{where } m(t) = \int_0^t \lambda(t')\,dt'$$

(c) The above pmf holds for $\lambda(t)$ sufficiently smooth that they can be approximated by a sequence of piecewise constant functions.

9.40

$$\begin{aligned}
P[N=k] &= \int_0^\infty P[N=k|T=t] f_T(t)\,dt \\
&= \int_0^\infty \frac{(\lambda t)^k}{k!} e^{-\lambda t} \beta e^{-\beta t}\,dt \\
&= \frac{\beta}{k!} \int_0^\infty (\lambda t)^k e^{-(\lambda+\beta)t}\,dt \qquad t' = (\lambda+\beta)t \\
&= \frac{\beta}{k!} \frac{\lambda^k}{(\lambda+\beta)^{k+1}} \underbrace{\int_0^\infty t'^k e^{-t'}\,dt'}_{\Gamma(k+1)=k!} \\
&= \left(\frac{\beta}{\lambda+\beta}\right)\left(\frac{\lambda}{\lambda+\beta}\right)^k \qquad k=0,1,\ldots
\end{aligned}$$

9.41

**6.37** $D(t) = N_1(t) - N_2(t)$ can assume negative values
$\Rightarrow D(t)$ cannot be a Poisson process.

(b)

The times between events in this process is the sum of two exponential random variables (Erlang). Thus the process is no longer Poisson.

9.42

**6.38 a)**

$$P[N_1(t) = j, N_2(t) = k | N(t) = k + j] = P[j \text{ heads in } k + j \text{ tosses}]$$
$$= \binom{j+k}{j} p^j (1-p)^k$$

b)

$$\begin{aligned} P[N_1(t) = j, N_2(t) = k] &= P[N_1(t) = j, N_2(t) = k, N(t) = k + j] \\ &= P[N_1(t) = j, N_2(t) = k | N(t) = k + j] P[N(t) = k + j] \\ &= \binom{j+k}{j} p^j (1-p)^k \frac{\lambda^{k+j}}{(k+j)!} e^{-\lambda} \\ &= \frac{(p\lambda)^j}{j!} e^{-p\lambda} \frac{((1-p)\lambda)^k}{k!} e^{-(1-p)\lambda} \end{aligned}$$

9.43

$$X(t) = \sum_{i=1}^{N(t)} X_i$$

$$P[X(t)=j] = \sum_{n=0}^{\infty} P[X(t)=j \mid N(t)=n]\, \underbrace{P[N(t)=n]}_{\frac{(\lambda t)^n}{n!} e^{-\lambda t}}$$

=

(a) $X_i$ is Bernoulli, then

$$P[X(t)=j \mid N(t)=n] = \binom{n}{j} p^j (1-p)^{n-j}$$

Proceeding as in prob. 9.35,

$$P[X(t)=j] = \frac{(\lambda t p)^j}{j!} e^{-\lambda t p}$$

(b) $X_i \in \{0, 5\}$ with $p_0 = \frac{5}{6}$ $p_1 = \frac{1}{6}$

same as (a):

$$P[X(t)=5j] = \frac{(\lambda t p)^j}{j!} e^{-\lambda t p} \qquad p = \frac{1}{6}.$$

(c) $X_i$ is Poisson with $\alpha = 1$, $X(t)$ is sum of $n$ iid Poisson w. mean $n$

$$P[X(t)=j \mid N(t)=n] = \frac{n^j}{j!} e^{-n}$$

$$P[X(t)=j] = \sum_{n=0}^{\infty} \frac{n^j}{j!} e^{-n} \frac{(\lambda t)^n}{n!} e^{-\lambda t}$$

$$= \frac{e^{-\lambda t + \frac{\lambda t}{e}}}{j!} \sum_{n=0}^{\infty} \frac{n^j \left(\frac{\lambda t}{e}\right)^n e^{-\frac{\lambda t}{e}}}{n!}$$

$E[N^j]$ for Poisson w rate $\lambda t/e$

(d)

$P[X(t)=j \mid N(t)=n]$

↑ there have been $n$ arrivals

↑ # of consecutive "tails" since last "heads" up to maximum of $n$ trials

$$\Rightarrow P[X(t)=j \mid N(t)=n] = \frac{(1-p)p^j}{(1-p^{n+1})} \qquad j=1,2,\ldots,n$$

$$P[X(t)=j] = \sum_{n=j}^{\infty} \frac{(1-p)p^j}{1-p^n} \frac{(\lambda t)^n}{n!} e^{-\lambda t}$$

after system has been in operation a long time (i.e. $t \to \infty$, $\lambda t$ is large and $p^n$ becomes negligible)

then

$$P[X(t)=j] \to (1-p)p^j \quad \text{geometric RV}$$

9.44

$Y(t)$ is a random telegraph process with transition rate $p\alpha$, since if $T$ = time till next transition, then

$$T = \tau_1 + \ldots + \tau_N \qquad \text{where } N \text{ is geometric RV}$$

$$\phi_T(N) = \mathcal{E}[\mathcal{E}[e^{j\omega t}|N]] = \mathcal{E}\left[\left(\frac{\alpha}{\alpha - j\omega}\right)^N\right] = \sum_{k=1}^{\infty}\left(\frac{\alpha}{\alpha - j\omega}\right)^k p(1-p)^{k-1}$$

$$= \frac{\alpha p}{\alpha - j\omega}\frac{1}{1-\frac{\alpha(1-p)}{\alpha - j\omega}} = \frac{\alpha p}{\alpha p - j\omega} \Rightarrow T \text{ exp. rate } \alpha p$$

$$\therefore P[Y(t)=+1] = \frac{1}{2} = P[Y(t)=-1] \text{ if } P[Y(0)] = \frac{1}{2}$$

and

$$C_Y(t_1,t_2) = e^{-2\alpha t|t_2-t_1|}$$

9.45

**6.41** Let $X(t)$ be the random telegraph process, then

$$\begin{aligned}
P[Y(t)=1] &= P[X(t)=1]=\frac{1}{2}\\
P[Y(t)=0] &= P[X(t)=-1]=\frac{1}{2}\\
m_Y(t) &= 1\left(\frac{1}{2}\right)+0\left(\frac{1}{2}\right)=\frac{1}{2}\\
C_Y(t_1,t_2) &= \mathcal{E}[Y(t_1)Y(t_2)]-\left(\frac{1}{2}\right)^2\\
&= P[Y(t_1)=1,Y(t_2)=1]-\frac{1}{4}\\
&= P[Y(t_1)=1]P[\text{even \# transitions in } t_2-t_1]-\frac{1}{4}\\
&= \frac{1}{2}(1+e^{-2\alpha|t_2-t_1|})\frac{1}{2}-\frac{1}{4}\\
&= \frac{1}{4}e^{-2\alpha|t_2-t_1|}
\end{aligned}$$

9.46

**6.42** a) $P[Z(t)=0|Z(0)=0]=$

$$\begin{aligned}
P[\text{even \# transitions in } [0,t]] &= \sum_{j=0}^{\infty}\frac{1}{1+\alpha t}\left(\frac{\alpha t}{1+\alpha t}\right)^{2j}\\
&= \frac{1}{1+\alpha t}\frac{1}{1-\left(\frac{\alpha t}{1+\alpha t}\right)^2}=\frac{1+\alpha t}{1+2\alpha t}
\end{aligned}$$

$P[Z(t)=0|Z(0)=1]=$

$$P[\text{odd \# transitions in } [0,t]]=\sum_{j=0}^{\infty}\frac{1}{1+\alpha t}\left(\frac{\alpha t}{1+\alpha t}\right)^{2j+1}=\frac{\alpha t}{1+2\alpha t}$$

$$\begin{aligned}
P[Z(t)=0] &= P[Z(t)=0|Z(t)=0]P[Z(0)=0]+P[Z(t)=0|Z(0)=1]P[Z(0)=1]\\
&= \frac{1+\alpha t}{1+2\alpha t}\frac{1}{2}+\frac{\alpha t}{1+2\alpha t}\frac{1}{2}=\frac{1}{2}\\
&\quad \text{where we assume } P[Z(0)=0]=\tfrac{1}{2}\\
P[Z(t)=1] &= 1-P[Z(t)=0]=\frac{1}{2}
\end{aligned}$$

b) $m_Z(t)=1\cdot P[Z(t)=1]=\frac{1}{2}$

9.47

~~6.43~~ $n(t) = u(t) - u(t-T)$

**a)** $$X(t) = \sum_{i=1}^{\infty} h(t-S_i) = \underbrace{\sum_{i=1}^{\infty} u(t-S_i)}_{\substack{\#\text{ arrivals to} \\ \text{time } t \\ N(t)}} - \underbrace{\sum_{i=1}^{\infty} u(t-T-S_i)}_{\substack{\#\text{ arrivals to} \\ \text{time } t-T \\ N(t-T)}}$$

$$\underbrace{\qquad\qquad\qquad\qquad\qquad\qquad}_{\text{increments in } (t-T,t)}$$

**b)** $\mathcal{E}[X(t)] = \lambda T$

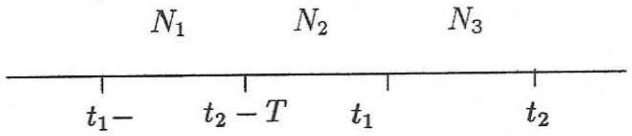
Assume $t_1 < t_2$

Case 1: If $t_1 < t_2 - T$




Then $\mathcal{E}[X(t_1)X(t_2)] = \mathcal{E}[X(t_1)]\mathcal{E}[X(t_2)] = \lambda^2 T^2$

Case 2: If $t_2 - T < t_1 < t_2$




Then

$$\begin{aligned}
\mathcal{E}[X(t_1)X(t_2)] &= \mathcal{E}[(N_1+N_2)(N_2+N_3)] \qquad \begin{matrix} X(t_1) = N_1 + N_2 \\ X(t_2) = N_2 + N_3 \end{matrix} \\
&= \mathcal{E}[N_1]\mathcal{E}[N_2] + \mathcal{E}[N_2^2] + \mathcal{E}[N_1]\mathcal{E}[N_3] + \mathcal{E}[N_2]\mathcal{E}[N_3] \\
\mathcal{E}[N_1] &= \lambda(t_2 - t_1) = \mathcal{E}[N_3] \\
\mathcal{E}[N_2] &= \lambda(t_1 - t_2 + T) \\
\mathcal{E}[N_2^2] &= VAR[N_2] + \mathcal{E}[N_2]^2 \\
&= \lambda(t_1 - t_2 + T) + \lambda^2(t_1 - t_2 + T)^2
\end{aligned}$$

$$\begin{aligned}
\therefore \mathcal{E}[X(t_1)X(t_2)] &= \lambda^2(t_1 - t_2 + T)^2 + \lambda(t_1 - t_2 + T) \\
&\quad + 2\lambda^2(t_2 - t_1)(t_1 - t_2 + T) + \lambda^2(t_2 - t_1)^2 \\
&= \lambda^2 T^2 + \lambda(t_1 - t_2 + T)
\end{aligned}$$

$$\Rightarrow C_X(t_1,t_2) = \begin{cases} \lambda(T - |t_2 - t_2) & |t_2 - t_1| < T \\ 0 & \text{otherwise} \end{cases}$$

9.48

~~6.44~~ a)

$$\begin{aligned}
\mathcal{E}[X^2(t)] &= \mathcal{E}[\mathcal{E}[X^2(t)/N(t)]] \\
\mathcal{E}[X^2(t)|N(t)=k] &= \mathcal{E}\left[\sum_{j=1}^{k}\sum_{i=1}^{k} h(t-S_j)h(t-S_i)\right] \\
&= k\mathcal{E}[h^2(t-s)] + k(k-1)\mathcal{E}[h(t-s)]^2 \\
\mathcal{E}[X^2(t)] &= \mathcal{E}[N(t)]\mathcal{E}[h^2(t-s)] + \mathcal{E}[N(t)(N(t)-1)]\mathcal{E}[h(t-s)]^2 \\
VAR[X(t)] &= \mathcal{E}[X^2(t)] - \mathcal{E}[X(t)]^2 \\
VAR[X(t)] &= \mathcal{E}[N(t)]\mathcal{E}[h^2(t-s)] + (\mathcal{E}[N^2] - \mathcal{E}[N])\mathcal{E}[h(t-s)]^2 \\
&\quad -\mathcal{E}[N]^2\mathcal{E}[h(t-s)]^2 \\
&= \underbrace{\mathcal{E}[N]}_{\lambda t}\mathcal{E}[h^2(t-s)] \\
&\quad + \underbrace{(\underbrace{VAR[N]}_{\lambda t} - \underbrace{\mathcal{E}[N]}_{\lambda t})}_{0}\mathcal{E}[h(t-s)]^2 \\
&= \lambda t\int_0^t h^2(t-s)\frac{ds}{t} \\
&= \lambda\int_0^t h^2(u)du
\end{aligned}$$

b)

$$\begin{aligned}
\mathcal{E}[X(t)] &= \lambda\int_0^t e^{-\beta u}du = \frac{\lambda}{\beta}(1-e^{-\beta t}) \to \frac{\lambda}{\beta} \\
VAR[X(t)] &= \lambda\int_0^t e^{-2\beta u}du = \frac{\lambda}{2\beta}(1-e^{-2\beta t}) \to \frac{\lambda}{2\beta}
\end{aligned}$$

9.49

Suppose $N = k$, the # of arrivals in 1 hr, and let $S_1, S_2, ..., S_k$ be the arrival times.



Then customer $i$ waits $60 - S_i$ minutes

$$\mathcal{E}[W|N=k] = \mathcal{E}\left[\sum_{i=1}^{k}(60-S_i)\right] = \sum_{j=1}^{k}\int_0^{60}(60-s)\frac{ds}{60} = 30k$$

Since arrival times are iid unif. dist. in [9.60]

$$\therefore \mathcal{E}[W] = \mathcal{E}[\mathcal{E}[W?N=k]] = \mathcal{E}[30N] = 30\mathcal{E}[N] = 30\lambda$$

## 9.5 Gaussian Random Processes, Wiener Process and Brownian Motion

9.50

$X(t)$ and $Y(t)$ independent

$\Rightarrow$ $X(t)$ and $Y(t)$ uncorrelated (whether Gaussian or not)

$X(t)$ and $Y(t)$ jointly Gaussian and uncorrelated

$\Rightarrow$ $X(t)$ and $Y(t)$ independent

(since joint pdf factors into product of marginal pdf's).

$X(t)$ and $Y(t)$ uncorrelated then

$X(t)$ and $Y(t)$ orthogonal if $m_X(t) = 0$ for all $t$ or $m_Y(t) = 0$ for all $t$.

$X(t)$ and $Y(t)$ orthogonal then

$X(t)$ and $Y(t)$ uncorrelated if $m_X(t) = 0$ for all $t$ or $m_Y(t) = 0$ for all $t$.

For jointly Gaussian RP's:

$X(t_1), Y(t_2)$ independent

$\Updownarrow$

$X(t_1), Y(t_2)$ uncorrelated

$\Updownarrow$ and $m_X(t) = 0$ or $m_Y(t) = 0$.

$X(t_1), Y(t_2)$ orthogonal and $m_X(t) = 0$ or $m_Y(t) = 0$.

9.51

$C_X(t_1,t_2) = 4e^{-2|t_1-t_2|}$ $X(t)$ & $X(t+s)$ are jointly Gaussian RVs with $COV(X(t),X(t+s)) = C_X(t,t+s) = 4e^{-2|s|}$

then $f_{X(t),X(t+s)}(x_1,x_2)$ can be achieved using Eq. 9.47 as follows:

$$f_{X(t),X(t+s)}(x_1,x_2) = \frac{e^{-\frac{1}{2}(x-m)^T K^{-1}(x-m)}}{2\pi |K|^{1/2}}$$

in which $m = 0$ $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$, $K = \begin{bmatrix} 4 & 4e^{-2|s|} \\ 4e^{-2|s|} & 4 \end{bmatrix}$

and $K^{-1} = \frac{1}{16(1-e^{-4|s|})}\begin{bmatrix} 4 & -4e^{-2|s|} \\ -4e^{-2|s|} & 4 \end{bmatrix}$ $|K| = 16(1-e^{-4|s|})$

So

$$f_{X(t),X(t+s)}(x_1,x_2) = \frac{e^{\left\{-\frac{1}{32(1-e^{-4|s|})}(4x_1^2+4x_2^2-8e^{-4|s|}x_1x_2)\right\}}}{8\pi\sqrt{1-e^{-4|s|}}}$$

9.52

$Z = 2tX - Y$ $X$ & $Y$ are jointly Gaussian, then $Z(t)$ would be a Gaussian RV too. (Example 6.24)

$m_{Z(t)} = E[Z(t)] = 2t\,E[X] - E[Y] = 2tm_X - m_Y$

$\sigma^2_{Z(t)} = VAR[Z(t)] = 4t^2\sigma_X^2 + \sigma_Y^2 + 4t\sigma_X\sigma_Y\rho_{XY}$

$$f_{Z(t)}(z) = \frac{e^{-\frac{(z-m_{Z(t)})^2}{2\sigma^2_{Z(t)}}}}{\sqrt{2\pi}\,\sigma_{Z(t)}}$$

9.53

(a) $E[Y(t)] = E[X(t+d)] - E[X(t)] = m_X(t+d) - m_X(t) = m_Y(t)$

$$E[Y(t_1)Y(t_2)] = E\left[\left(X(t_1+d) - X(t_1)\right)\left(X(t_2+d) - X(t_2)\right)\right]$$

$$= E[X(t_1+d)X(t_2+d)] - E[X(t_1)X(t_2+d)] - E[X(t_2)X(t_1+d)] + E[X(t_1)X(t_2)]$$

$$C_Y(t_1,t_2) = E[Y(t_1)Y(t_2)] - m_Y(t_1)\,m_Y(t_2)$$

$$= C_X(t_1+d,t_2+d) - C_X(t_1,t_2+d) - C_X(t_2,t_1+d) + C_X(t_1,t_2)$$

(b) Since $X(t)$ and $X(t+d)$ are jointly Gaussian RV's, then $Y(t)$ is also Gaussian with mean $m_Y(t)$ and

$$\sigma_Y^2(t) = C_Y(t,t) = C_X(t+d,t+d) - 2C_X(t,t+d) + C_X(t,t)$$

(c) $Y(t)$ and $Y(t+s)$ are jointly Gaussian RVs defined by the linear transformation of jointly Gaussian RV's

$$Y(t) = X(t+d) - X(t)$$

$$Y(t+s) = X(t+s+d) - X(t+s)$$

$\therefore$ $Y(t)$ and $Y(t+s)$ have a jointly Gaussian pdf with mean $m_Y(t)$, variance $\sigma_Y^2(t)$ and covariance $C_Y(t_1,t_2)$.

(d) Any $n$ time samples of $Y(t)$ are defined by a linear transformation of jointly Gaussian random variables. Thus the samples have a jointly Gaussian pdf.

9.54

**6.16** a) $\mathcal{E}[X(t)] = \mathcal{E}[A\cos\omega t + B\sin\omega t] = \mathcal{E}[A]\cos\omega t + \mathcal{E}[B]\sin\omega t = 0$

$$\begin{aligned} C_X(t_1,t_2) &= \mathcal{E}[X(t_1)X(t_2)] - m_X(t_1)m_X(t_2) \\ &= \mathcal{E}[(A\cos\omega t_1 + B\sin\omega t_1)(A\cos\omega t_2 + B\sin\omega t_2)] \\ &= \mathcal{E}[A^2]\cos\omega t_1\cos\omega t_2 + \mathcal{E}[A]\mathcal{E}[B]\cos\omega t_1\sin\omega t_2 \\ &\quad + \mathcal{E}[A]\mathcal{E}[B]\cos\omega t_2\sin\omega t_1 + \mathcal{E}[B^2]\sin\omega t_2\sin\omega t_2 \\ &= \sigma^2(\cos\omega t_1\cos\omega t_2 + \sin\omega t_1\sin\omega t_2) \\ &= \sigma^2\cos\omega(t_1 - t_2) \end{aligned}$$

b) Because $A$ and $B$ are jointly Gaussian RV's, $X(t) = A\cos\omega t + B\sin\omega t$ and $X(t+d) = A\cos\omega(t+s) + B\sin\omega(t+d)$ are also jointly Gaussian, with zero means and covariance matrix

$$K = \begin{bmatrix} \sigma^2 & \sigma^2\cos\omega d \\ \sigma^2\cos\omega d & \sigma^2 \end{bmatrix} \qquad |K|^{1/2} = \sigma^2|1 - \cos^2\omega d| = \sigma^2|\sin\omega d|$$

$$K^{-1} = \frac{1}{\sigma^4\sin^2\omega d}\begin{bmatrix} \sigma^2 & -\sigma^2\cos\omega d \\ -\sigma^2\cos\omega d & \sigma^2 \end{bmatrix}$$

$$\begin{aligned} f_{X(t)X(t+d)}(x_1,x_2) &= \frac{\exp\left\{-\frac{1}{2}\underline{x}'K^{-1}\underline{x}\right\}}{2\pi\sigma^2|\sin\omega s|} \\ &= \frac{\exp\left\{-\frac{x_1^2 - 2\cos\omega d\, x_1x_2 + x_2^2}{2\sigma^2\sin^2\omega d}\right\}}{2\pi\sigma^2|\sin\omega d|} \end{aligned}$$

9.55

**6.18** $Z(t) = X(t)\cos\omega t + Y(t)\sin\omega t$

a) $\mathcal{E}[Z(t)] = m_X(t)\cos\omega t + m_Y(t)\sin\omega t = 0$

$$\begin{aligned} C_Z(t_1,t_2) &= \mathcal{E}[(X(t_1)\cos\omega t_1 + Y(t_1)\sin\omega t_1) \\ &\quad (X(t_2)\cos\omega t_2 + Y(t_2)\sin\omega t_2)] \\ &= \mathcal{E}[X(t_1)X(t_2)]\cos\omega t_1\cos\omega t_2 \\ &\quad + \underbrace{\mathcal{E}[X(t_1)Y(t_2)]}_{0}\cos\omega t_1\sin\omega t_2 \\ &\quad + \underbrace{\mathcal{E}[Y(t_1)X(t_2)]}_{0}\sin\omega t_1\cos\omega t_2 + \mathcal{E}[Y(t_1)Y(t_2)]\sin\omega t_1\sin\omega t_2 \\ &= C(t_1,t_2)\cos\omega t_1\cos\omega t_2 + C(t_1,t_2)\sin\omega t_1\sin\omega t_2 \\ &= C(t_1,t_2)\cos\omega(t_1 - t_2) \end{aligned}$$

b) $f_{Z(t)}(z) = \dfrac{1}{\sqrt{2\pi C(t,t)}}e^{-z^4/2C(t,t)}$

9.56

**6.19** $\mathcal{E}[Y(t)] = \mathcal{E}[X^2(t)] = C_X(t,t)$

$\mathcal{E}[Y(t_1)Y(t_2)] = \mathcal{E}[X^2(t_1)X^2(t_2)]$

To proceed further we need the result from Problem 4.69

$$\begin{aligned}\mathcal{E}[X^2(t_1)X^2(t_2)] &= \mathcal{E}[X^2(t_1)]\mathcal{E}[X^2(t_2)] + 2\mathcal{E}[X(t_1)X(t_2)]^2 \\ \Rightarrow \mathcal{E}[Y(t_1)Y(t_2)] &= C_X(t_1,t_1)C_X(t_2,t_2) + 2C_X^2(t_1,t_2)\end{aligned}$$

Since arrival times are iid unif. dist. in [9.60]

$$\therefore \mathcal{E}[W] = \mathcal{E}[\mathcal{E}[W?N=k]] = \mathcal{E}[30N] = 30\mathcal{E}[N] = 30\lambda$$

9.57

**9.46 a)**
$$\begin{aligned}F_{Y(t)}(y) &= P[Y(t) \le y] = P[X(t) + \mu t \le y] = P[X(t) \le y - \mu t] \\ &= F_{X(t)}(y - \mu t) \\ \Rightarrow f_{Y(t)}(y) &= F_X(y - \mu t) = \frac{1}{\sqrt{2\pi\alpha t}} e^{-(y-\mu t)^2/2\alpha t}\end{aligned}$$

**b)**
$$\begin{aligned}F_{Y(t)Y(t+s)}(y_1,y_2) &= P[X(t) + \mu t \le y_1, X(t+s) + \mu(t+s) \le y_2] \\ &= F_{X(t),X(t+s)}(y_1 - \mu t, y_2 - \mu(t+s)) \\ \Rightarrow f_{Y(t)Y(t+s)}(y_1,y_2) &= f_{X(t),X(t+s)}(y_1 - \mu t, y_2 - \mu(t+s)) \\ &= f_{X(t)}(y_1 - \mu t) f_{X(s)}(y_2 - y_1 - \mu s) \\ &= \frac{3^{-(y_1-\mu t)^2/2\alpha t}}{\sqrt{2\pi\alpha t}} \frac{e^{-(y_2-y_1-\mu s)^2/2\alpha s}}{\sqrt{2\pi\alpha s}}\end{aligned}$$

9.58

~~6.47~~ **a)** $Y(t) = X^2(t)$

$$\Rightarrow f_{Y(t)}(y) = \frac{1}{\sqrt{2\pi\alpha t y}} e^{-y/2\alpha t} \text{ from Example } \cancel{3.26}\ 4.33$$

**b)** $f_{Y(t_2)}(y_2|Y(t_1) = y_1) = \dfrac{f_{Y(t_2),Y(t_1)}(y_2, y_1)}{f_{Y(t_1)}(y_1)}$

$$\begin{aligned} Y(t_1) &= X^2(t_1) \\ Y(t_2) &= X^2(t_2) \end{aligned}$$

$$\begin{aligned} J(x(t_1), x(t_2)) &= \begin{vmatrix} 2x(t_1) & 0 \\ 0 & 2x(t_2) \end{vmatrix} = 4|x(t_1)x(t_2)| \\ &= 4\sqrt{y(t_1)y(t_2)} \end{aligned}$$

$$\begin{aligned} f_{Y(t_1)Y(t_2)}(y_1, y_2) &= \frac{f_{X_1X_2}(\sqrt{y_1}, \sqrt{y_2})}{4\sqrt{y_1y_2}} + \frac{f_{X_1X_2}(\sqrt{y_1}, -\sqrt{y_2})}{4\sqrt{y_1y_2}} \\ &\quad + \frac{f_{X_1X_2}(-\sqrt{y_1}, \sqrt{y_2})}{4\sqrt{y_1y_2}} + \frac{f_{X_1X_2}(-\sqrt{y_1}, -\sqrt{y_2})}{4\sqrt{y_1y_2}} \\ &= \frac{e^{-\frac{y_1}{2\alpha t_1}} \left\{ e^{-\frac{y_2+y_1+2\sqrt{y_1y_2}}{2\alpha(t_2-t_1)}} + e^{-\frac{y_2+y_1+2\sqrt{y_1y_2}}{2\alpha(t_2-t_1)}} \right\}}{4\pi\alpha\sqrt{t_1(t_2-t_1)y_1y_2}} \end{aligned}$$

$$\Rightarrow f_{Y(t_2)}(y_2|Y(t_1) = y_1) = \frac{e^{-\frac{y_2+y_1-2\sqrt{y_1y_2}}{2\alpha(t_2-t_1)}} + e^{-\frac{y_2+y_1+2\sqrt{y_1y_2}}{2\alpha(t_2-t_1)}}}{2\sqrt{2\pi\alpha(t_2-t_1)y_2}}$$

9.59

~~6.48~~ 6.24 a) From Example ~~4.32~~ we know that $Z(t) = X(t) - \alpha X(t-s)$ is a Gaussian RV since $X(t)$ and $X(t-s)$ are jointly Gaussian. Therefore we need only find $m_Z(t)$ and $VAR[Z(t)]$

$$\begin{aligned}
m_Z(t) &= \mathcal{E}[X(t)] - a\mathcal{E}[X(t-s)] = 0 \\
VAR[Z(t)] &= \mathcal{E}[(X(t) - aX(t-s))^2] \\
&= \mathcal{E}[X^2(t)] - 2a\mathcal{E}[X(t)X(t-s)] + a^2\mathcal{E}[X^2(t-s)] \\
VAR[Z(t)] &= \alpha t - 2a(\alpha(t-s)) + a^2\alpha(t-s) \\
&= \alpha t(1 - 2a + a^2) + 2a\alpha s - a^2\alpha s \\
&= \alpha t(a-1)^2 - a\alpha s(a-2) \\
f_{Z(t)}(z) &= \frac{\exp\left\{-\frac{z^2}{2VAR[Z(t)]}\right\}}{\sqrt{2\pi VAR[Z(t)]}}
\end{aligned}$$

b)
$$\begin{aligned}
m_Z(t) &= E[X(t) - aX(t-s)] = 0 \\
C_Z(t_1, t_2) &= E[\{Z(t_1) - m(t_2)\}\{Z(t_2) - m(t_2)\}] \\
&= E[\{X(t_2) - aX(t_1 - s)\}\{X(t_2) - aX(t_2 - s)\}] \\
&= \alpha \min(t_1, t_2) - a\alpha \min(t_1 - s, t_2) \\
&\quad -a\alpha \min(t_1, t_2 - s) + a^2\alpha \min(t_1, t_2)
\end{aligned}$$

9.60 $\alpha = 1$ & $0 < t < 1$

a) Using Eq. 9.52 we have:

$$f_{X(t),X(1)}(x_1,x_2) = \frac{\exp\left\{-\frac{1}{2}\left[\frac{x_1^2}{t} + \frac{(x_2-x_1)^2}{(1-t)}\right]\right\}}{2\pi\sqrt{t(1-t)}}$$

b) $$f_{X(t)}(x \mid X(0)=X(1)=0) = f_{X(t),X(1)}(x_1,0) = \frac{\exp\left\{-\frac{1}{2}\frac{x_1^2}{t(1-t)}\right\}}{2\pi\sqrt{t(1-t)}}$$

c) $$f_{X(t)}(x \mid X(t_1)=a, X(t_2)=b) = f_{Y(t)}(y \mid Y(t_2-t_1)=b-a)$$

$$= f_{Y(t),Y(t_2-t_1)}(y, b-a)$$

$$= \frac{\exp\left\{-\frac{1}{2}\left[\frac{y^2}{t} + \frac{(b-a-y)^2}{(t_2-t_1-t)}\right]\right\}}{2\pi\sqrt{t(t_2-t_1-t)}}$$

End of Section: END OF PART I.